



Matheson Constructors Limited
245 Industrial Pkwy North
Aurora, Ontario L4G 4C4
P: (905) 669-7999
F: 905 669-0268

Project: 2025-048 Docksteader PRPS Reporting
Station
6 Docksteader Rd.
Brampton, Ontario L6R

Submittal #08 11 13-2.0 - Hollow metal Frame shop drawings 08 11 13 - STEEL DOORS AND FRAMES

Revision	0	Submittal Manager	Alvin Hernandez (Matheson Constructors Limited)
Status	Open	Date Created	Apr 21, 2025
Issue Date		Spec Section	08 11 13 - STEEL DOORS AND FRAMES
Responsible Contractor	Egress Systems of Canada Ltd.	Received From	Scott Minhinnick (Egress Systems of Canada Ltd.)
Received Date		Submit By	Nov 17, 2025
Final Due Date	May 20, 2026	Lead Time	84 day(s)
		Cost Code	
Location		Type	Shop drawing
Submittal Package			
Approvers	Alvin Hernandez (Matheson Constructors Limited), Ashish Kumar (Diamond Schmitt Architects)		
Ball in Court	Alvin Hernandez (Matheson Constructors Limited)		
Distribution	Ali Haidar (Matheson Constructors Limited), Alvin Hernandez (Matheson Constructors Limited), Kathy Schaeffer (Matheson Constructors Limited), Marwan Diab (Matheson Constructors Limited), Dhruv Patel (Matheson Constructors Limited), Hari Singh (Matheson Constructors Limited)		
Description	Enclosed fore your review and acceptance is Submittal 08 11 13-02-R0-HM Door and Frames Shop Dwgs		
Schedule ID #			

Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					
Scott Minhinnick		May 6, 2026	May 6, 2026	Submitted	Submittal 08 11 13-02-R0-HM Door and Frames Shop Dwgs.pdf
Alvin Hernandez	May 6, 2026	May 6, 2026	May 6, 2026	Reviewed as noted	
Ashish Kumar	May 6, 2026	May 20, 2026	Jun 11, 2026	Reviewed as noted	304 08 11 13-2.0. Hollow metal Frame shop drawings_RN.pdf (Current) Transmittal_Submittal - 304 (Responded and Closed).pdf (Current)
Comment	DSA(MS): Revise and Resubmit all elements that are flagged for clarity and records.				

Diamond Schmitt | 384 Adelaide Street West, Suite 100 Toronto ON M5V 1R7 Canada

PROJECT: Dockstader PRPS 221033 DATE SENT: 6/11/2026

SUBJECT: Hollow metal Frame shop drawings SUBMITTAL ID: 304

TYPE: Submittal TRANSMITTAL ID: 02253

PURPOSE: Reviewed as Noted VIA: Procore

SPEC SECTION: 08 11 13

FROM

NAME	COMPANY	EMAIL	PHONE
Ashish Kumar	Diamond Schmitt	akumar@dsai.ca	1-416-862-8800

TO

NAME	COMPANY	EMAIL	PHONE
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Ali Haidar	Matheson Constructors	ahaidar@mathesonconstructors.com	
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Tristan Crawford	Diamond Schmitt	TCrawford@dsai.ca	1-416-862-8800 ext. 641

Submittal Transmittal

DATE: 6/11/2026
ID: 02253

REMARKS: DSA(MS): Revise and Resubmit all elements that are flagged for clarity and records.

CONTENTS

QUANTITY:	1	DATED:	5/6/2026	NUMBER:
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DESCRIPTION:

08 11 13-2-0 - Submittal Form.pdf

ACTION:

REMARKS:

QUANTITY:	1	DATED:	5/6/2026	NUMBER:
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DESCRIPTION:

Submittal 08 11 13-02-R0-HM Door and Frames Shop Dwgs.pdf

ACTION:

REMARKS:

COPIES:

Ajit Stephen	(Region of Peel Deveolpment Services - Planning)
Majadeh Hamidsaeidi	(Region of Peel Deveolpment Services - Planning)
Jean-Louis Bramwell	(Total Opening Consultants)
Thennarasu Parimalam	(Total Opening Consultants)



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Project: 2025-048 Docksteader PRPS Reporting Station
6 Docksteader Rd.
Brampton, Ontario L6R

**Submittal #08 11 13-2.0 - Hollow metal Frame shop drawings
08 11 13 - STEEL DOORS AND FRAMES**

Revision	0	Submittal Manager	Lizet Pabon (Matheson Constructors Limited)
Status	Open	Date Created	Apr 21, 2025
Issue Date		Spec Section	08 11 13 - STEEL DOORS AND FRAMES
Responsible Contractor		Received From	
Received Date		Submit By	Nov 17, 2025
Final Due Date	May 20, 2026	Lead Time	84 day(s)
		Cost Code	
Location		Type	Shop Drawing
Submittal Package			
Approvers	Alvin Hernandez (Matheson Constructors Limited), Ashish Kumar (Diamond Schmitt Architects)		
Ball in Court	Ashish Kumar (Diamond Schmitt Architects)		
Distribution	Ali Haidar (Matheson Constructors Limited), Alvin Hernandez (Matheson Constructors Limited), Kathy Schaeffer (Matheson Constructors Limited), Marwan Diab (Matheson Constructors Limited), Dhruv Patel (Matheson Constructors Limited), Hari Singh (Matheson Constructors Limited)		
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Submittal Workflow

Name	Sent Date	Due Date	Returned Date	Response	Attachments
General Information Attachments					
Scott Minhinnick		May 6, 2026	May 6, 2026	Submitted	Submittal 08 11 13-02-R0-HM Door and Frames Shop Dwgs.pdf
Alvin Hernandez	May 6, 2026	May 6, 2026	May 6, 2026	Reviewed as noted	
Ashish Kumar	May 6, 2026	May 20, 2026		Pending	

245 Industrial Parkway North
Aurora, ON L4G 4C4
Phone: (905) 669-7999
Fax: 905-669-0268

Date: May 6, 2026 **Job No.:** 2025-048

To: Diamond Schmitt

Attn: Ashish Kumal

C.C.: Project File, A. Stephens, M. Sinno, K. Schaeffer, M. Diab, A. Haidar, H. Singh, D. Patel, A. Hernandez,

Re: Submittal 08 11 13-02-R0-HM Door and Frames Shop Dwgs

WE ARE SENDING YOU 1 electronic copy via email of the following items:

ZERO CARBON DOCUMENTS	SHOP DRAWINGS	PRODUCT INFORMATION	SAMPLES / MOCK UP	SUBSTITUTION REQUEST	WARRANTY		
		X					

NO.	DATE	REV #	DESCRIPTION
1	May 6/26	0	Dwg. 1 of 12: General Notes, Abbreviations and Schedules
2	May 6/26	0	Dwg. 2 to 5 of 12: Hollow Metal Frame and Door Schedule
3	May 6/26	0	Dwg. 6 of 12: Door Elevations
4	May 6/26	0	Dwg. 7 of 12: Frame Elevations
5	May 6/26	0	Dwg. 8 of 12: Screens
6	May 6/26	0	Dwg. 9 of 12: Screens
7	May 6/26	0	Dwg. 10 of 12: Profiles
8	May 6/26	0	Dwg. 11 of 12: Sections
9	May 6/26	0	Dwg. 12 of 12: Anchorage
10	May 6/26	0	Guide Specification For Installation and Storage of Hollow Metal Frames.

THESE ARE TRANSMITTED as checked below:

☒ For approval
☐ For signature
☐ For your information
☐ As requested
☐ For review and comment

REMARKS:

Alvin Hernandez

Signature

This transmission, including any accompanying attachments, contains information that may be privileged and/or exempt from disclosure under applicable law, and is intended only for the use of the recipient(s) named above. Any distribution, review, dissemination, or copying of the contents of this communication by anyone other than the intended recipient(s) is strictly prohibited. If you receive this communication in error, please notify the sender immediately and destroy the copy you have received. Thank you.

diamond schmitt			
In 2026/05/06		Out 2026-06-11	
Project No. 221033		Specification Section No. 08 11 13	
Reviewed	Reviewed as Noted MS	Revise and Resubmit	Not Reviewed
This review by Diamond and Schmitt Architects Incorporated is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of any engineering elements which form part of the contract documents prepared by others. This review shall not mean that Diamond and Schmitt Architects Incorporated approves the detailed design inherent in the submittal, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the submittal or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.			

DSA(MS): Revise and Resubmit all elements that are flagged for clarity and records.

		TOTAL OPENING CONSULTANTS
SHOP DRAWING REVIEW		
This review is for the sole purpose of ascertaining conformance with the general design concept. The review shall not mean the detail design inherent in the shop drawings is approved. The review shall not relieve the Contractor of their responsibility for errors or omissions in the shop drawings or their responsibility for meeting requirements of the contract documents.		
☐	REVIEWED	
☒	REVIEWED AS NOTED	
☐	REVISE AND RESUBMIT	
BY: TP		DATE: 05-07-2026

	ARCH #	
RFI 003	D1A-STB	THESE FRAMES CAN NOT BE FIRE RATED AT 350MM JAMB DEPTH MAX JAMB DEPTH 330MM
	D3A-STD	THE MAXIMUM ALLOWABLE JAMB DEPTH IS 330MM
	D1A-STD	Frames for doors D1A-STB , D3A-STD , D1A-STD , D107 , D2A-STB , D2A-STD , D1B-STD, and D131B to be revised from "F2" type frames to F5 type frames as per SI-003. To accommodate jamb size of 330mm and new 5mm GL14 glass (GL14 glass revised under SI-017)
	D107	
	D2A-STB	
	D2A-STD	
RFI 003	D1A-STB	DOORS CAN NOT BE FIRE RATED WITH 13MM GLASS. THE MAX GLASS THICKNESS 10MM
	D3A-STD	
	D1A-STD	Doors can receive 13mm glass with interlocking seam construction
	D107	
	D2A-STB	
	D2A-STD	
	D131B	
RFI 008	D220	PLEASE CONFIRM HOLLOW METAL FRAME # D220 TO BUT UP TO SC02 ALUMINUM SCREEN.

Door D220: Frame is to be revised from 'F2' to 'F4' as per SI-017

MATERIALS

Commercial grade steel to ASTM A653, CS, Type B, Coating Designation ZF75 (A40) Galvanneal

Galvanised G90 Material hot dipped ZF275 coating

Meeting all requirements of CSDMA

Minimum steel thicknesses in accordance with the Architects specification and Appendix 1 of the CSDMA

ZINC PRIMER APPLIED AT ALL DOOR, FRAMES AND SCREENS AT ALL WELDED AND GRINDED AREAS

GLASS AND GLAZING NOTES.

ALL DOORS, FRAMES, AND SCREENS ARE MANUFACTURED AS PER SPECIFICATION DIVISION 08 11 13
INTERIOR AND EXTERIOR SCREENS OR TRANSOM FRAMES TO BE GLAZED WITH ADJUSTABLE GLASS STOPS UP TO 25mm
PENDING FIRE RATINGS AND MAXIMUM GLASS SPACE AND THICKNESS REQUIREMENTS.
ALL SCREWS FOR THE GLAZING COME COMPLETE WITH HOLLOW METAL PRODUCTS.

PLEASE FOLLOW METAL DOOR SHOP DRAWINGS FOR GLASS SIZE THICKNESS AND LOCATION OF GLASS SIDE.
METAL DOOR LTD. IS NOT RESPONSIBLE FOR INCORRECT GLASS INSTALLATION, LOST SCREWS OR GLAZING STOPS.

Recognized By



Canadian Steel Door
Manufacturers Association
L'Association Canadienne
Des Fabricants de Portes D'Acier



Toronto
Construction
Association
Since 1827
Proudly Supporting Member



Canada



Intertek



FOR MORE INFORMATION ON ALL HOLLOW METAL PRODUCTS, REQUIREMENTS AND SPECIFICATIONS

PLEASE VISIT WWW.CSDMA.ORG

EGRESS
SYSTEMS OF CANADA LTD

ES25-015-002

Consultant: this submittal is to be reviewed by Architect, Hardware.

Consultant: Refer to Submittal 08 11 13-01-R1-SR-009-Revised HM Doors and Frames(RAN)

Subcontractor:

- Prep doors in accordance with the hardware schedule
- Prep Door frames to receive required junction boxes, conduits, etc.
- Coordinate glazing components with the Glazing Subcontractor.

Dockstader

6 Dockstader Road, Br

File No. SP/

diamond
schmitt

MATHESON

EGRESS
SYSTEMS OF CANADA LTD

Missing:
A) D131.1

B)D133.1A: Refer to PC-017

C)D302D,

D) D202C,

MATHESON CONSTRUCTORS

Reviewed
Reviewed as Noted
Revise and Resubmit
Not Reviewed

Date: 2026-05-06

Reviewed by:

This review by MATHESON CONSTRUCTORS LIMITED is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of engineering elements which form part of the contract documents prepared by others. This review shall not mean that MATHESON CONSTRUCTORS LIMITED approves the detailed design inherent in the shop drawings, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the shop drawings or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.

ABBREVIATIONS

W/C	ZF120 (A40) GALVANNEAL
G90	ZF275 HOP DIPPED G90 GALVANIZED
HM	HOLLOW METAL
WD	WOOD
BL	BORROWED LITE
FR	FRAME
SC	SCREEN
PR	PAIR
L	LEFT
R	RIGHT
DA	DOUBLE ACTING
T	THERMALLY BROKEN
HT	HIGH TRAFFIC
EXT	EXTERIOR
INT	INTERIOR
HS	HOLLOW STEEL
HC	HONEY COMB

HOLLOW METAL TOLERANCES IS +/- 1.5 MM
ANY REQUIRED SITE MEASUREMENTS WILL BE
THE RESPONSIBILITY OF THE CONTRACTOR.

DATE:

MARCH 20 2026

DRAWN BY

DJL

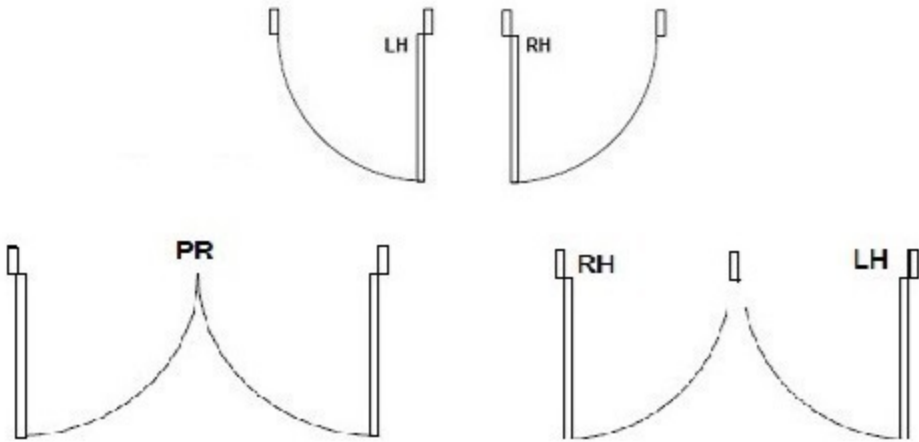
DIVISION 8
08 11 13

SUBMISSION

4

S.O. # 45012
ES25-015-002

SWINGS



METAL DOOR LTD



HOLLOW METAL FRAME AND DOOR SCHEDULE

Consultant: Please advise on the comments provided by Egress

Job No: 45012

Revision Date: 20 Mar 2026 9:54

LOCATION				DOOR INFO										FRAME INFO													Revision Date: 20 Mar 2026 9:54						
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS						PEMKO		ARCH			MD ELEV	PEMKO		THERMAL BREAK		WALL		REMARKS				
											SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R	SWING	TYPE	FINISH	Ga		SECTIONS	RAIN DRIP	WD	HT	THCK	ANCH					
1	D1A-STA	SC	1	RFI 007	J	FG	HM	45	HT	19	3/4	L	1065	x	2440	FG	W/C	16			L	SC08	W/C	16	S1		15.E.G			1065 X 2440		T/A	SPRINKLER PROTECTED
2	D1A-STB	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F5	W/C	16	A		ADVISE			965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
3	D1A-STD	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	R	965	x	2440	VL	W/C	16			R	F5	W/C	16	A		ADVISE			965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
4	D1B-STD	FR	1	13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A		17			965 X 2440		S/F	DSA: Door frame to align with poured Conc. wall at both these locations to maintain fire rating
5	D101C	FR	1	TOC: A?	A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
6	D107	FR	1	RFI 003 13MM GLASS	A	VL	HM	45	HT	19	1HR	L	965	x	2440	VL	W/C	16			L	F5	W/C	16	A		ADVISE			965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
7	D108	FR	1	TOC: A?	A		WD	45	INT	19		L	965	x	2440						L	F5	W/C	16	A		17			965 X 2440		S/F	DSA: Door frame to align with poured Conc. wall at D107 to maintain fire rating. Additional revised F5 detail forthcoming to clarify.
8	D109	FR	1		A		WD	45	INT	19		L	965	x	2440						L	F2	W/C	16	A		6			965 X 2440		C/S	
9	D116E	FR	1	13MM GLASS	C	VL	HM	45	HT	19		R	965	x	2440	VL	W/C	16			R	F2	W/C	16	A		10			965 X 2440		T/A	
10	D118C	FR	1	13MM GLASS	C	VL	HM	45	HT	19		L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
11	D119A	FR	1		C		WD	45	INT	19		R	965	x	2440						R	F2	W/C	16	A		13			965 X 2440		C/S	
12	D119B	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		13			965 X 2440		C/S	DSA: At all locations where wood door is indicated, Finish to be WDVN-1 - Ref. Spec section 06 40 00.2.4 and finish schedule A053
13	D121	FR	1		A	F	HM	45	HT	19	0HR	L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
14	D122	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
15	D123	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
16	D125A	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	
17	D125B	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
18	D126A	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
19	D126B	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16			R	F2	W/C	16	A		8			965 X 2440		T/A	DSA: Door frame to align with poured Conc. wall at D2A-STB to maintain fire rating. Additional revised F5 detail forthcoming to clarify.
20	D127A	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
21	D130D	FR	1	13MM GLASS	D	G	HM	45	HT	19		L	965	x	2440						L	F2	W/C	16	A		8			965 X 2440		T/A	
22	D131B	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	1HR	L	965	x	2440	VL	W/C	16			L	F2	W/C	16	A		8			965 X 2440		T/A	
24	D2A-STB	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F5	W/C	16	A		ADVISE			965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
25	D2A-STD	FR	1	RFI 003 13MM GLASS	C	VL	HM	45	HT	19	3/4	L	965	x	2440	VL	W/C	16			L	F5	W/C	16	A		ADVISE			965 X 2440		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
26	D2A-STE	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		13			965 X 2440		C/S	DSA: Door frame to align with poured Conc. wall at D2A-STD to maintain fire rating
27	D204	FR	1		A		WD	45	HT	19		L	965	x	2440						L	F5	W/C	16	A		17			965 X 2440		S/F	
28	D205	FR	1		A		WD	45	HT	19		L	965	x	2440						L	F5	W/C	16	A		17			965 X 2440		S/F	
29	D206	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16			L	F2	W/C	16	A		12			965 X 2440		C/S	
30	D207A	FR	1		L	F1	HM	45	HT	19	3/4	R	965	x	2440	F1	W/C	16			R	F2	W/C	16	A		12			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR CHECK HARDWARE FOR CUTOUT SIZES
31	D207B	FR	1		L	F1	HM	45	HT	19	3/4	L	965	x	2440	F1	W/C	16			L	F2	W/C	16	A		13			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR CHECK HARDWARE FOR CUTOUT SIZES
32	D208	FR	1		A		WD	45	HT	19		L	965	x	2440						L	F2	W/C	16	A		12			965 X 2440		C/S	
33	D209	FR	1		A		WD	45	INT	19		R	965	x	2440						R	F2	W/C	16	A		12			965 X 2440		C/S	
34	D213B	FR	1	212 TO 213	A		WD	45	HT	19		R	1220	x	2440						R	F2	W/C	16	A2		13F			1220 X 2440		C/S	OPEN CASE SLIDER FRAME
35	D215	FR	1		A		WD	45	HT	19		R	965	x	2440						R	F2	W/C	16	A		13			965 X 2440		C/S	
36	D215B	FR	1	WASH TO 215	A		WD	45	INT	19		R	915	x	2440						R	F2	W/C	16	A2		12F			915 X 2440		C/S	OPEN CASE SLIDER FRAME
37	D216	FR	1		A		WD	45	INT	19		R	965	x	2440						R	F2	W/C	16	A		13			965 X 2440		C/S	

Door D220: Door frame type to be revised to 'F4" type frame. Refer to SI-017

HOLLOW METAL FRAME AND DOOR SCHEDULE

Job No: 45012

Revision Date: 20 Mar 2026 9:54

	LOCATION				DOOR INFO										FRAME INFO												Revision Date: 20 Mar 2026 9:54					
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS						PEMKO		ARCH SWING	TYPE	FINISH	Ga	MD ELEV	PEMKO		THERMAL BREAK		WALL		REMARKS		
											SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R	SECTIONS						RAIN DRIP	WD	HT	THCK	ANCH				
38	D217	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		11			965 X 2440		C/S	
39	D219	FR	1		A	F1	HM	45	HT	19	0HR	L	965	x	2440	F1	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S	LOUVRE CUTOUT IN DOOR. CHECK HARDWARE FOR CUTOUT SIZES
40	D220	FR	1	RFI 008	C		WD	45	INT	19		L	965	x	2440					L	F4	W/C	16	A		13			965 X 2440		*	*C/S HINGE SIDE STRAP ANCHOR STRIKE SIDE. PLEASE CONFIRM ELEVATION
41	D221	FR	1		D		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S	
42	D222	FR	1		A		WD	45	INT	19		L	965	x	2440					L	F2	W/C	16	A		13			965 X 2440		C/S	
43	D223	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		13			1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16														
44	D224	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		13			965 X 2440		C/S	
45	D225	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		6			1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16														
46	D227	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	
47	D229A	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		PR	F2	W/C	16	B		13			1930 X 2440		C/S	
												R	965	x	2440	F	W/C	16														
48	D229C	FR	1		A	F	HM	45	HT	19	3/4	L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	
49	D230	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S	
50	D231	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S	
51	D232	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		13			965 X 2440		C/S	
52	D233	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		13			965 X 2440		C/S	
53	D235	FR	1		A		WD	45	INT	19		R	965	x	2440					R	F2	W/C	16	A		12			965 X 2440		C/S	
54	D236	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S	
55	D237	FR	1		A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		12			965 X 2440		C/S	
56	D238	FR	1		A	F	HM	45	HT	19		L	965	x	2440	F	W/C	16		L	F2	W/C	16	A		12			965 X 2440		C/S	
57	D239	FR	1	TOC: A?	A	F	HM	45	HT	19		R	965	x	2440	F	W/C	16		R	F2	W/C	16	A		12			965 X 2440		C/S	
58	D3A-STD	FR	1	RFI 003 13MM GLASS	A	VL	HM	45	INT	19	3/4	R	965	x	2135	VL	W/C	18		R	F5	W/C	16	A		ADVISE			965 X 2135		S/F	FRAME CAN NOT BE FIRE RATED JAMB DEPTH CAN NOT EXCEED 330MM
59	D3B-STD	FR	1		A	F	HM	45	INT	19	3/4	R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		13			965 X 2135		T/A	
60	D302C	FR	1	TOC: D302D MISSING?	A	F	HM	45	HT	19	3/4	L	1067	x	2440	F	W/C	16		PR	F2	W/C	16	B		13			2134 X 2440		T/A	DSA: Door frame to align with poured Conc. wall at D3A-STD to maintain fire rating
												R	1067	x	2440	F	W/C	16														
61	D303	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		12			965 X 2135		C/S	
62	D304	FR	1		A	F	HM	45	INT	19		L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		15			965 X 2135		C/S	
63	D305	FR	1		A	F	HM	45	INT	19		R	965	x	2135	F	W/C	18		R	F2	W/C	16	A		12			965 X 2135		C/S	
64	D306	FR	1		A	F	HM	45	INT	19	3/4	L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		12			965 X 2135		C/S	
65	D307	FR	1		A	F	HM	45	INT	19	3/4	L	965	x	2135	F	W/C	18		L	F2	W/C	16	A		12			965 X 2135		C/S	
66	D1A-STE	FR	1		A	F	HM	45	EXT	19		L	965	x	2440	F	G90	16		L	F1	G90	16	A		T8	☒		965 X 2440		S/F	
67	D1C-STD	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	
68	D101B	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	
69	D102F	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	
70	D116A	FR	1	13MM GLASS	C	VL	HM	45	EXT	19		R	965	x	2440	VL	G90	16		R	F1	G90	16	A		T14	☒		965 X 2440		S/F	

HOLLOW METAL FRAME AND DOOR SCHEDULE																														Job No: 45012	
	LOCATION				DOOR INFO										FRAME INFO															Revision Date: 20 Mar 2026 9:54	
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT	THK	UND CUT	LABEL	DOOR LEAF DETAILS					PEMKO		ARCH SWING	TYPE	FINISH	Ga	MD ELEV	PEMKO SECTIONS	THERMAL BREAK		WALL		REMARKS			
											SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R							RAIN DRIP	WD	HT	THCK	ANCH			
71	D130A	FR	1	13MM GLASS	D	G	HM	45	EXT	19		L	965	x	2440	G	G90	16		L	F1	G90	16	A		T7	☒		965 X 2440		S/F
72	D131A	FR	1	13MM GLASS	C	VL	HM	45	EXT	19		L	965	x	2440	F	G90	16		L	F1	G90	16	A		T8	☒		965 X 2440		S/F
73	D138	FR	1		A	F	HM	45	EXT	19		R	965	x	2440	F	G90	16		R	F1	G90	16	A		15	☐	☐	965 X 2440		S/F
74	D229B	FR	1		A	F	HM	45	EXT	19		L	965	x	2135	F	G90	16		L	F1	G90	16	A		T14	☒	☐	965 X 2135		S/F
75	D3A-STB	FR	1		A	F	HM	45	EXT	19		L	965	x	2135	F	G90	16		L	F1	G90	16	A		14	☐	☐	965 X 2135		S/F
76	D302A	FR	1		A	F	HM	45	EXT	19		R	1065	x	2050	F	G90	16		R	F1	G90	16	A		T14	☒	☐	1065 X 2050		S/F
77	D302B	FR	1	TOC: A?	A	F	HM	45	EXT	19		R	1065	x	2135	F	G90	16		R	F1	G90	16	A		T7	☒	☐	1065 X 2135		S/F
78	RH-01	FR	1		A	F	HM	45	EXT	19		L	915	x	915	F	G90	16		L	A1	G90	16	A1		T18	☒	☐	915 X 915		S/F
81	SC09-17	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
82	SC09-16	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
83	SC09-15	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
84	SC09-14	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
85	SC09-13	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
86	SC09-12	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
87	SC09-11	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
88	SC09-1	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
89	SC09-9	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
90	SC09-18	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
91	SC09-7	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
92	SC09-6	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
93	SC09-5	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
94	SC09-4	SC	1						INT											SC09	W/C	16	S2		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
95	SC09-3	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
96	SC09-2	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
97	SC09-19	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
98	SC09-8	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
99	SC09-10	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
100	SC09-21	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
101	SC09-20	SC	1						INT											SC09	W/C	16	S2		8.E.G	☐	☐	SEE ELEV		T/A	SPRINKLER PROTECTED
102	SC10-13	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
103	SC10-12	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
104	SC10-11	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
105	SC10-10	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
106	SC10-9	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
107	SC10-1	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
108	SC10-8	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED
109	SC10-14	SC	1						INT											SC10	W/C	16	S3		6.E.G	☐	☐	SEE ELEV		C/S	SPRINKLER PROTECTED

TOC: 20 MIN FR?

TOC: A?

DSA: D138 & D3A-STB - NO FRR

DSA: Screens SC09, SC10, SC14 & SC15 to be Glass Type GL-11 - 13mm Laminated Tempered Glass, Clear

Print Date: Mar 20 2026

METAL DOOR LTD. P.O. Box 638, Gormley, Ontario. L0H 1G0 www.metaldoorltd.com

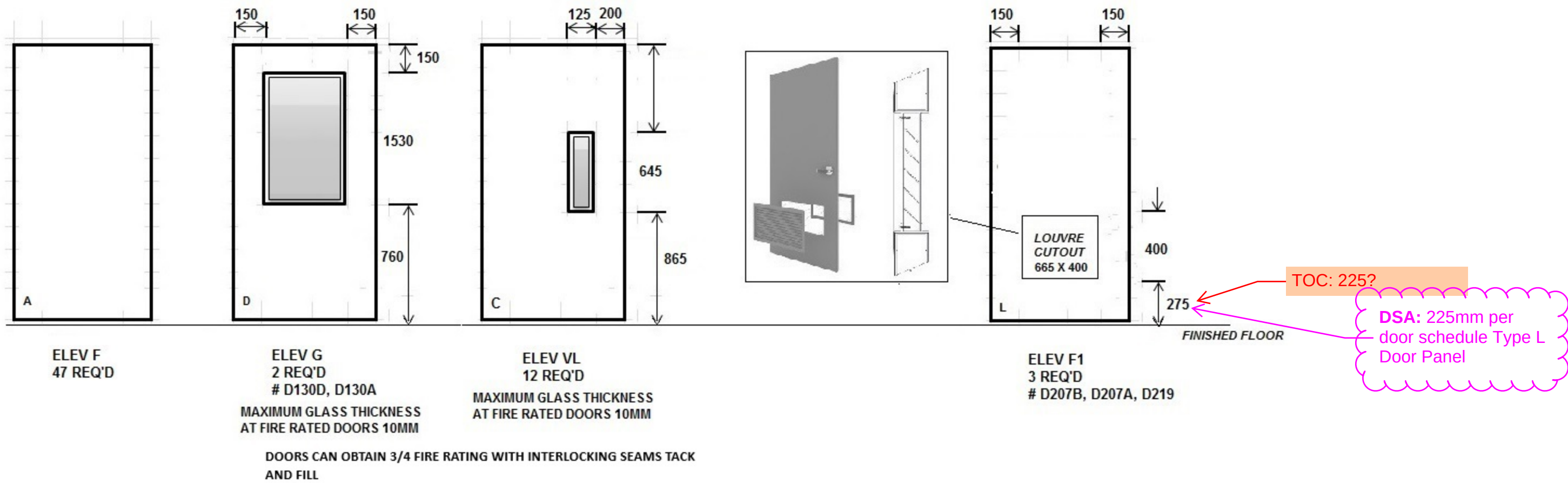
Page 4 of 4

HOLLOW METAL FRAME AND DOOR SCHEDULE																								Job No: 45012					
	LOCATION				DOOR INFO							FRAME INFO												Revision Date: 20 Mar 2026 9:54					
INDX	ARCH #	TYPE	PARTS	ITEM NOTES	ARCH TYPE	MD ELEV	MAT THK	UND CUT	LABEL	DOOR LEAF DETAILS			PEMKO		ARCH SWING	MD TYPE	PEMKO FINISH	MD Ga	PEMKO ELEV	THERMAL BREAK		WALL				REMARKS			
										SW	WIDTH	HEIGHT	TYPE	FINISH	Ga	T/R				SECTIONS	RAIN DRIP	WD	HT	THCK	ANCH				
110	SC10-7	SC	1						INT									SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
111	SC10-6	SC	1						INT									SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
112	SC10-5	SC	1						INT									SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
113	SC10-4	SC	1						INT									SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
114	SC10-3	SC	1						INT									SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
115	SC10-2	SC	1						INT									SC10	W/C	16	S3		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
116	SC14	SC	1						INT									SC14	W/C	16	S4		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED
117	SC15	SC	1						INT									SC15	W/C	16	S5		6.E.G			SEE ELEV		C/S	SPRINKLER PROTECTED

DSA: Screens SC09, SC10, SC14 & SC15 to be Glass Type GL-11 - 13mm Laminated Tempered Glass, Clear

DSA: Screens SC09, SC10, SC14 & SC15 to be Glass Type GL-11 - 13mm Laminated Tempered Glass, Clear

DSA: Screen SC15 to include GDRL6 in front - similar to what is shown at SC14 - proposed change forthcoming



INTERIOR DOORS
18 GA A40 GALVANNEAL
WELDED AND FILLED INTERLOCKING SEAMS GROUND SMOOTH
HONEYCOMB BOARD INSULATION

TACK AND FILL

WARNOCK-HERSEY FIRE RATED UL LISTED

INTERIOR SCREWED IN FIRE RATED GLASS STOPS
FOR RATED GLASS 7 TO 10MM THICK MAX.

EXTERIOR DOORS
16 GA ZF275 G90 GALVANIZED
WELDED LAP SEAM
STEEL STIFFENED
URETHANE FOAM INSULATION

FULLY WELDED

FLUSH STEEL TOP CAPS

EXTERIOR GLASS STOPS
REMOVABLE SCREW IN GLASS STOPS
UP TO 25MM NON RATED GLASS THICKNESS

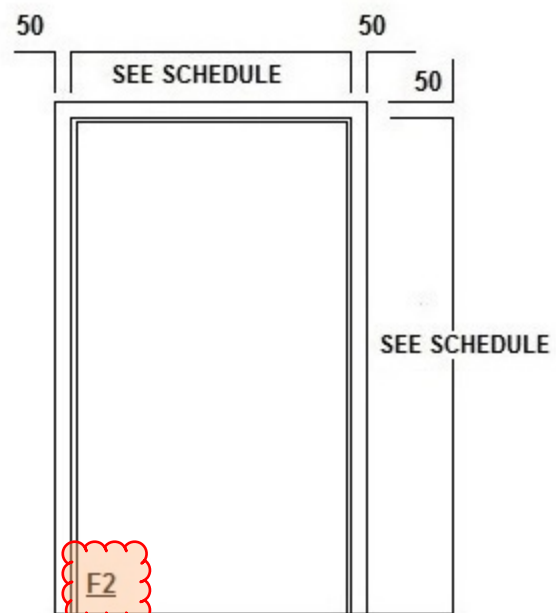
IT IS NOT RECOMMENDED TO PAINT EXTERIOR DOORS A DARK COLOR

HIGH TRAFFIC DOORS
16 GA A40 GALVANNEAL
FULLY WELDED LAP SEAM
STEEL STIFFENED
POLYISOCYANURATE CORE

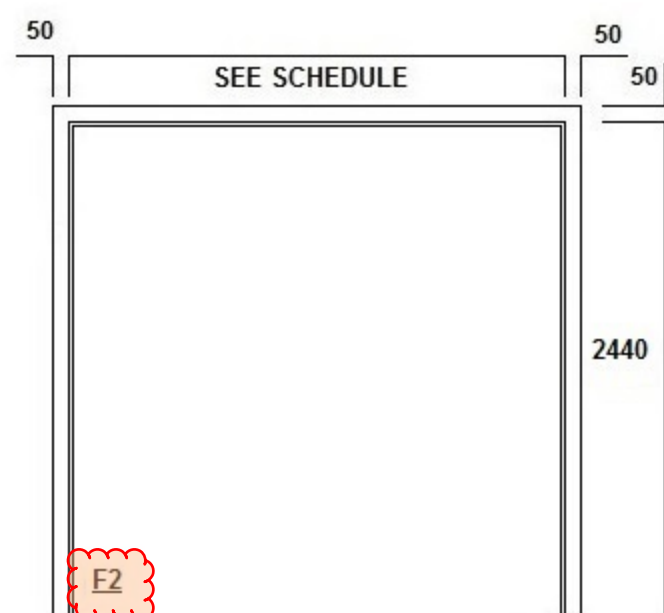
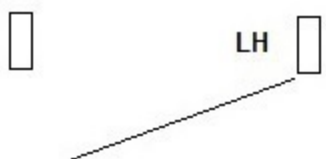
FULLY WELDED

WARNOCK-HERSEY FIRE RATED UL LISTED

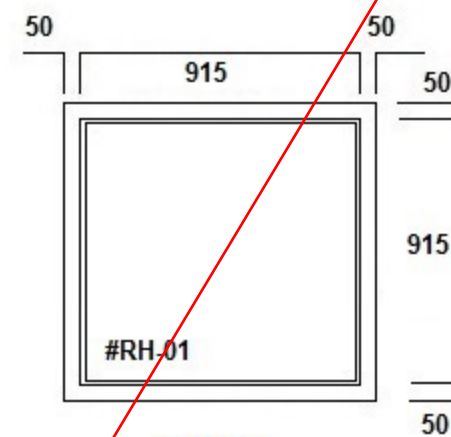
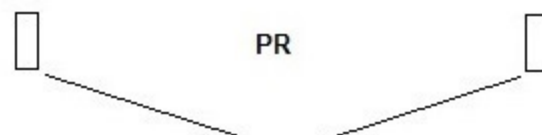
INTERIOR SCREWED IN FIRE RATED GLASS STOPS
FOR RATED GLASS 7 TO 10MM THICK MAX.



ELEV A
69 REQ'D



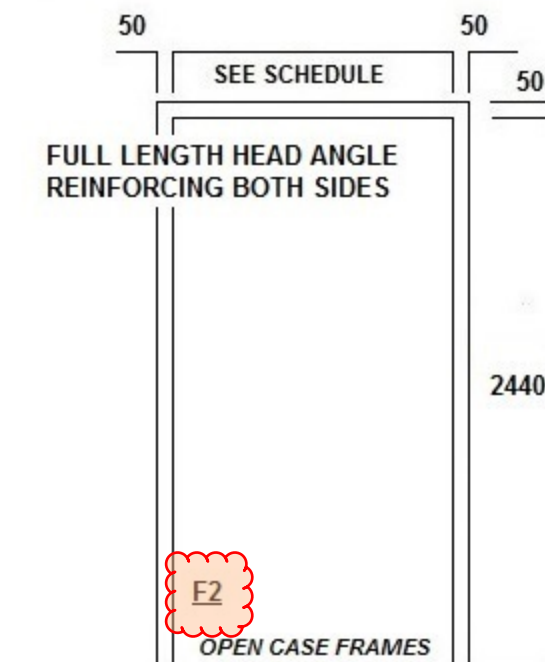
ELEV B
4 REQ'D



ELEV A1
1 REQ'D

4 SIDED ROOF HATCH
2 X S/F ON ALL 4 SIDES

By Roofer



ELEV A2
2 REQ'D

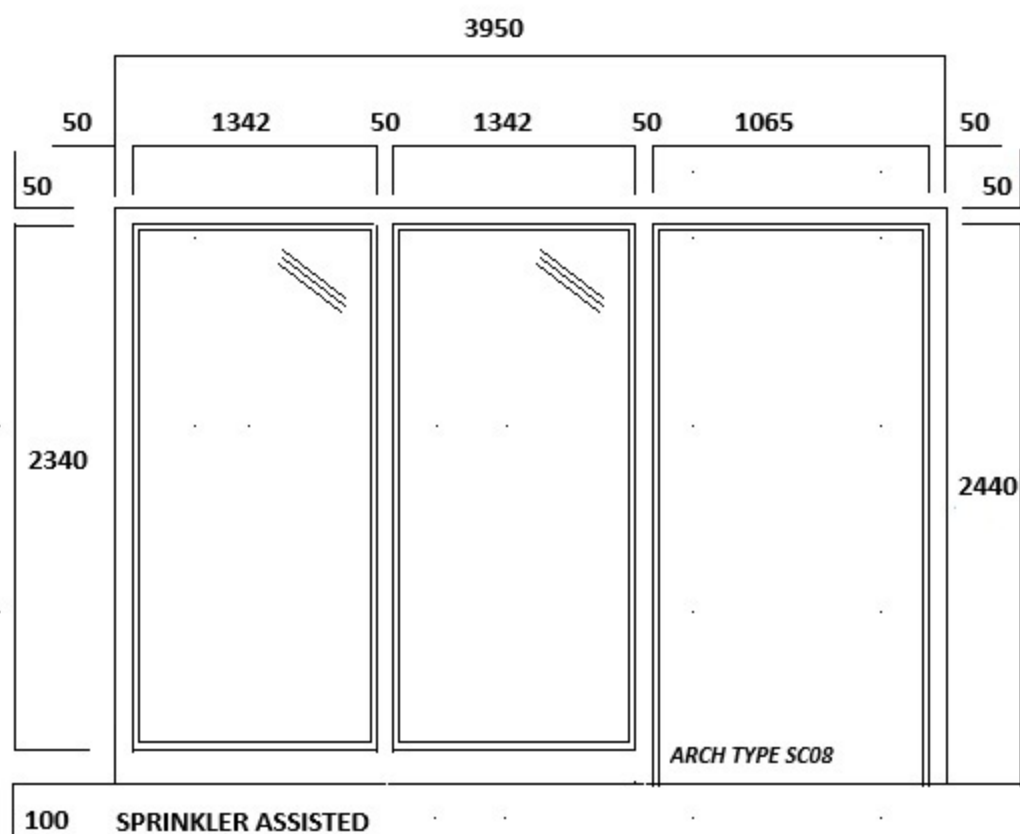


D213B, D215B

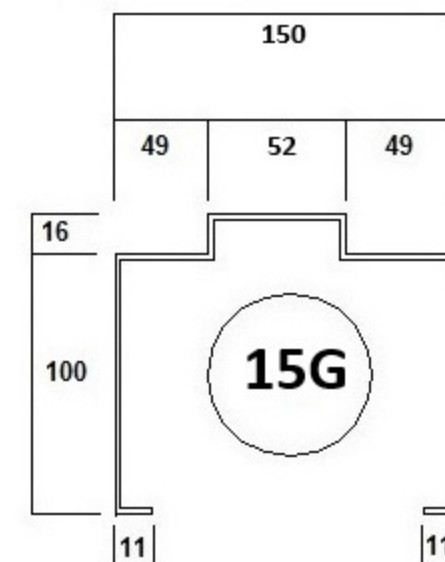
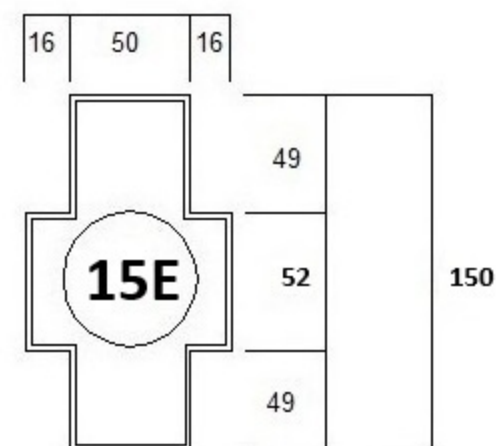
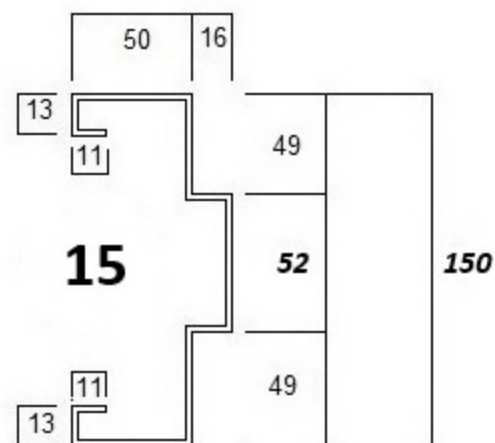
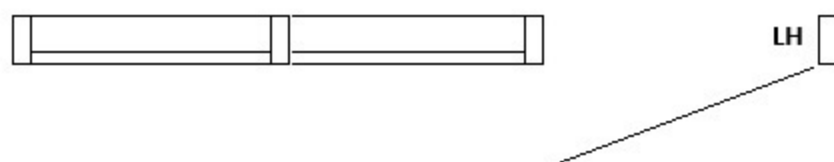
TOC:
F2 - 69+4+2=75 qty
A1 - 1 qty
sc08 - 1 qty
Total - 77 qty

But door schedule shows 78 line items. Also
F1 type is 12 no.s of the 78 shown. Clarify
exact qty of each type

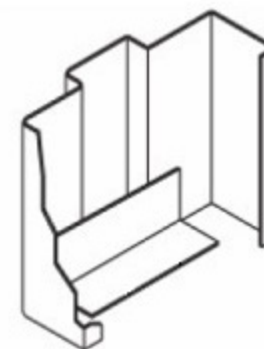
Egress to Advise



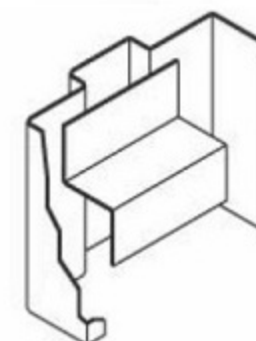
ELEV S1
1 REQ'D
D1A-STA



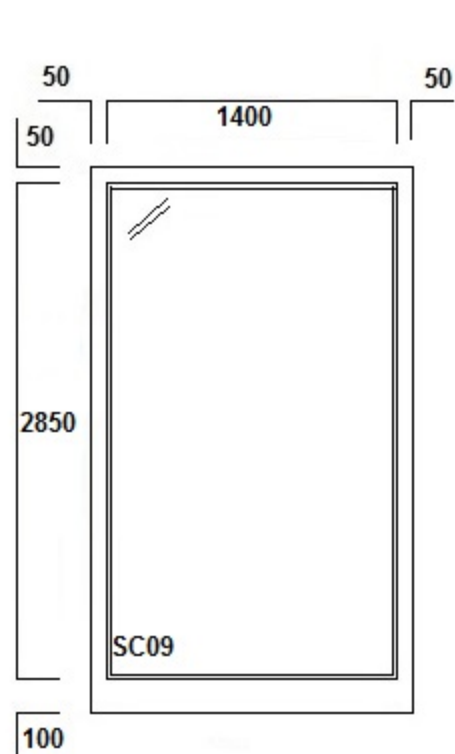
ADJUSTABLE MULLION BASE ANCHOR
SHIPPED IN MULLION OR LOOSE



WELDED FLOOR ANCHOR (T/A)

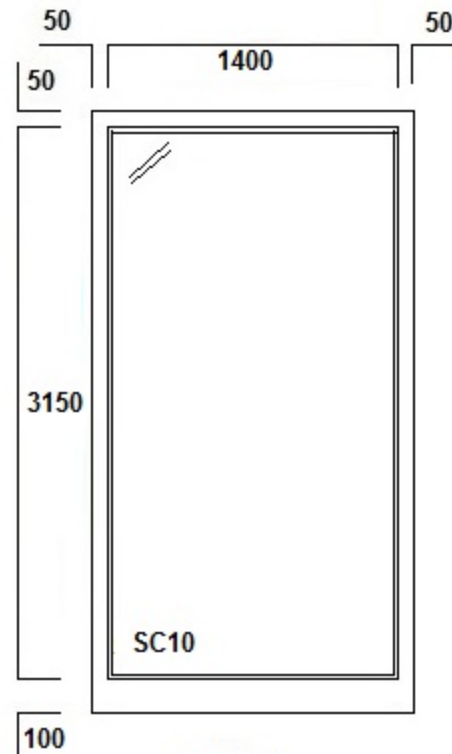


DRYWALL CHANNEL STUD ANCHOR (C/S)
WITH WELDED (T/A)



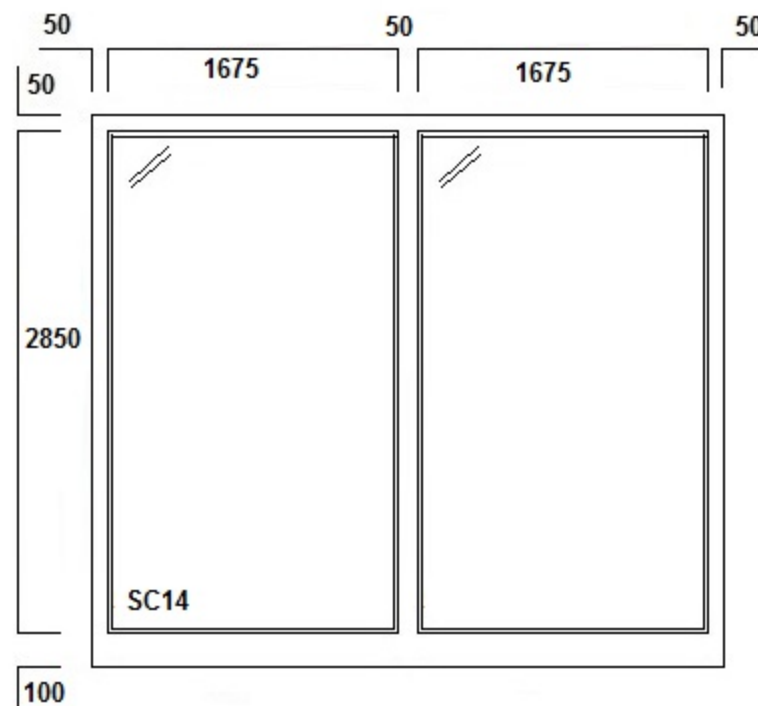
ELEV S2
21 REQ'D

SPRINKLER PROTECTED



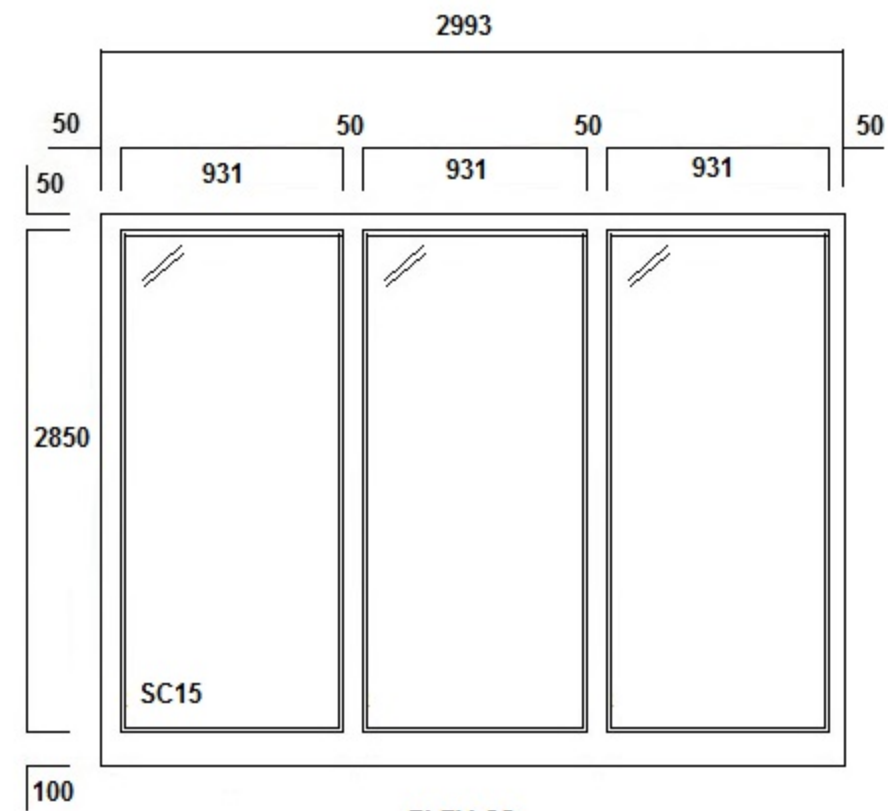
ELEV S3
14 REQ'D

SPRINKLER PROTECTED



ELEV S4
1 REQ'D

SPRINKLER PROTECTED



ELEV S5
1 REQ'D

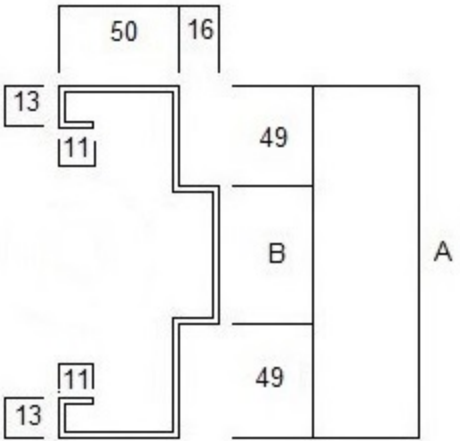
SPRINKLER PROTECTED

PLEASE CONFIRM ALL DIMENSIONS

NOT FOR REVIEW.

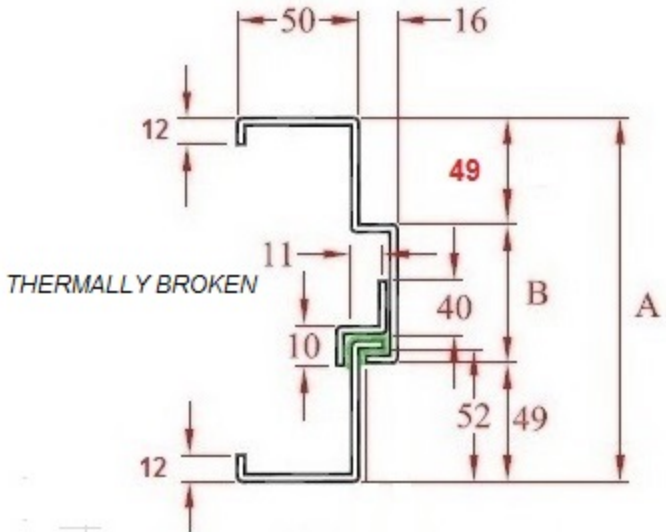


HOLLOW METAL SECTIONS



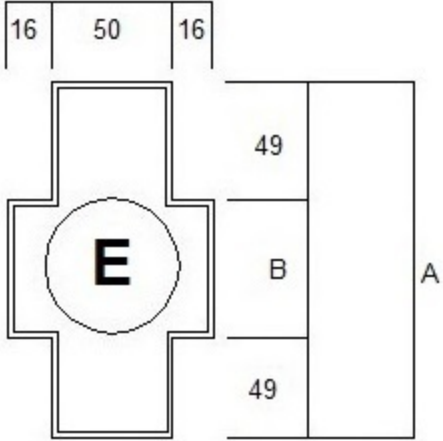
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	B	GIRTH FRONT	BRAKE LINE
12	124	26		
13	140	42		
14	145	47		
15	150	52		
6	156	58		
8	190	92		
10	240	142		
15	265	167		
16	330	232	MAX JAMB DEPTH THAT CAN BE FIRE RATED.	
17	350	252		



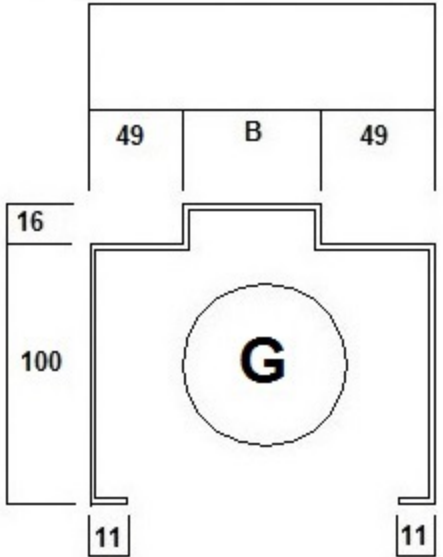
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	B	GIRTH FRONT	BRAKE LINE
14T	145	47		
7T	184	86		
8T	190	92		
18T	250	152		



"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

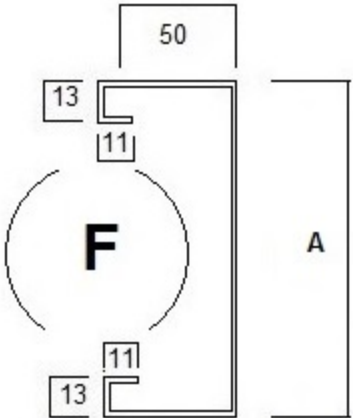
SECTION	A	B	GIRTH FRONT	BRAKE LINE
15E	150	52		
6E	156	58		
8E	190	92		



"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	B	GIRTH FRONT	BRAKE LINE
15G	150	52		
6G	156	58		
8G	190	92		

PLEASE CONFIRM ALL BASES TO BE 100mm

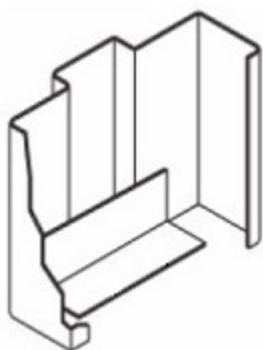


OPEN CASE FRAMES # A1-540A , A1-516A

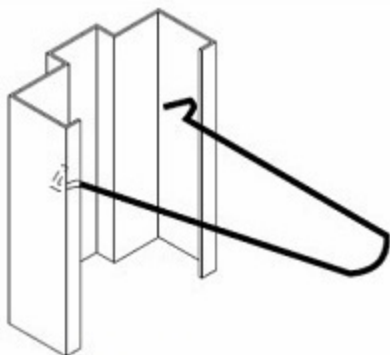
"A" DIMENSIONS AS PER ARCHITECTS DOOR SCHEDULE

SECTION	A	GIRTH FRONT	BRAKE LINE
12F	124		
13F	140		

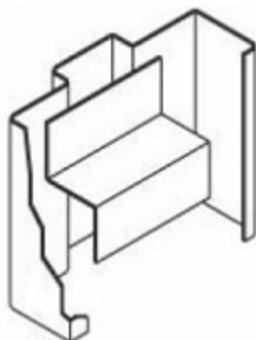
RECOMMENDED ANCHORAGE



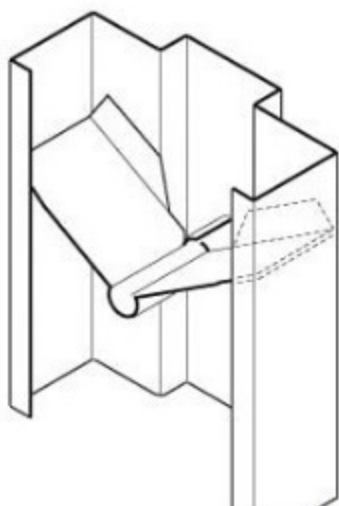
WELDED FLOOR ANCHOR (T/A)



MASONRY WIRE ANCHORS SHIPS LOOSE
WITH WELDED (T/A)



DRYWALL CHANNEL STUD ANCHOR (C/S)
WITH WELDED (T/A)



SCREW FIXED - EMA (S/F)
WITH GUIDE AND FRAME SUPPORT WELDED IN
3/8 FASTENERS SUPPLIED BY OTHERS



All Anchorage to be confirmed on shop drawings.
6 anchors per frame or screen up to 2286mm
8 anchors per frame or screen above 2286mm

Some anchors are shipped loose.

Extra or lost anchors can be ordered extra to contract.



METAL DOOR LTD

GUIDE SPECIFICATION FOR INSTALLATION AND STORAGE OF HOLLOW METAL DOORS AND FRAMES

MVC MATHESON CONSTRUCTORS

Reviewed	(✓)
Reviewed as Noted	()
Revise and Resubmit	()
Not Reviewed	()

Date: **2026-05-06**

Reviewed by:

This review by MATHESON CONSTRUCTORS LIMITED is for the sole purpose of ascertaining conformance with the general design concept for architectural features only, and does not in any way constitute review of the design of engineering elements which form part of the contract documents prepared by others. This review shall not mean that MATHESON CONSTRUCTORS LIMITED approves the detailed design inherent in the shop drawings, responsibility for which shall remain with the Contractor submitting same, and such review shall not relieve the Contractor of the responsibility for errors or omissions in the shop drawings or of the responsibility for meeting all requirements of the contract documents. The Contractor is responsible for dimensions to be confirmed and correlated at the job site, for information that pertains solely to fabrication processes or to techniques of construction and installation, and for coordination of the work of all trades.

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L'ASSOCIATION
CANADIENNE
DES FABRICANTS
DE PORTES D'ACIER

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STORAGE AND INSTALLATION OF HOLLOW METAL DOORS AND FRAMES

PART I - INTRODUCTION

This guide specification is intended to stress the necessary precautions and requirements for receipt, storage, handling and installation of hollow metal products, and the skills essential in the hanging of doors.

The proper performance of most products depends not only on how they are manufactured, but how they are installed. This is especially true of hollow metal doors and frames. The installation of doors and frames is an operation demanding care and skill, if the doors are to operate properly. Care in manufacturing does not, in itself, guarantee satisfactory performance. Even the best designed and most carefully constructed frames and doors, if incorrectly installed, will not function properly.

Hollow metal work is fabricated in accordance with the shop drawings, approved by the architect or engineer. Preparation for hardware or other items supplied by others is provided in accordance with the information furnished to the hollow metal manufacturer.

The hollow metal manufacturer is a material supplier, not a subcontractor. The manufacturer does not include the installation of their product in the building, but only shipment in good condition from the factory.

Should the General Contractor discover any error in the hollow metal delivered to the job site, it is imperative that the hollow metal manufacturer be notified in writing and allowed sufficient time before initiating any corrective measure in the field, so that the manufacturer can participate in solving the problem. Failure to do so could result in the cancellation of the warranty and/or fire label and non-acceptance of any cost associated with repair.

Most member companies of the Hollow Metal Manufacturer's Association Division of CSDMA have their own field representatives who are qualified not only to do expert repair work but to determine whether the fault lies with the manufacturer or with some other party.

It is essential that material is properly stored prior to installation and skills are exercised in the setting of frames and hanging of doors.

PART 2 - RECEIVING AND STORAGE OF MATERIAL

A. RECEIVING

Upon delivery, the contractor responsible for receiving hollow metal products shall thoroughly inspect for damage. Cardboard and other wrappings shall be removed for inspection and to promote air circulation. Any scratches or disfigurements caused in shipping or handling shall be promptly cleaned and touched up with a direct to metal rust inhibitive primer.

Should damaged material be found, the General Contractor has the option of refusing delivery or to accept the material as damaged. For coordination purposes, CSDMA suggests that delivery should not be refused, but rather accepted as damaged. Any damaged items should be noted on the freight bill. Claims will not be honored by the freight carrier, unless the damaged items are noted on the freight bill at the time of delivery. The contractor shall notify the hollow metal manufacturer in writing immediately of any item signed for as damaged.

B. ON SITE STORAGE

Proper storage of hollow metal work at the construction site will help prevent damage to the primer coat of paint. Prime coated steel must be protected when exposed to the elements such as high humidity, salt air, rain, snow, and/or damp wrappings etc.... Particular attention must, therefore, be given to steel products having a coat of factory applied primer. Primer is porous to properly receive and hold top coats. Water or moisture, in contact with primer coated steel will seep through to the steel. An electrolytic action then follows, resulting in corrosion and causing the paint film to lose adhesion. The presence of oxygen at the water-air interface behind the loosened paint film accelerates corrosive action and the prime coat further deteriorates.

Even when zinc coated steel is used to provide corrosion resistance, manufacturers of hollow metal door and frame products have found that one week of product exposure to water, due to improper storage, can be equivalent to at least a year of outdoor exposure to the elements.

NOTE: Paint manufacturers advise that the primer typically used by hollow metal manufacturers should receive a finish coat of paint within 30 days of delivery. It is the responsibility of the General Contractor to sand, touch up and clean prime painted surfaces prior to finish painting in accordance with the finish paint manufacturer's instructions.

The following procedures shall always be observed in storing hollow metal doors and frames at the job site:

1. Store all materials in a dry area, under cover. All products shall be stored where they will not be exposed to, or come in contact with the elements.
2. Do not use non-vented plastic or canvas. These materials create a humidity chamber, which promotes blistering and corrosion.
3. Store doors and frames in an upright position with heads uppermost, Figures 1 and 2.
4. Place no more than 5 doors or welded frames in a group. Small groups not only minimize the likelihood of damage due to excess handling, but also facilitate selection from the group for installation. In the case of multi-opening frames, no more than three units should be stored in a group, to avoid serious damage to the bottom most frame.
5. Place all material on planking or blocking at least 4 in. (100 mm) off the ground, 2 in. (50 mm) off a paved area or the floor slab.
6. Provide at least 1/4 in. (6.4 mm) space between all units to permit air circulation.

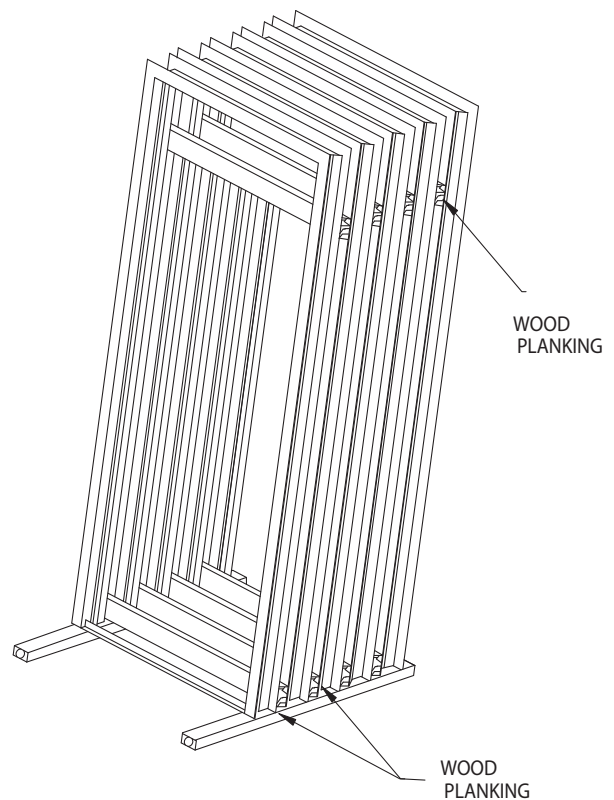


FIGURE 1
FRAME STORAGE

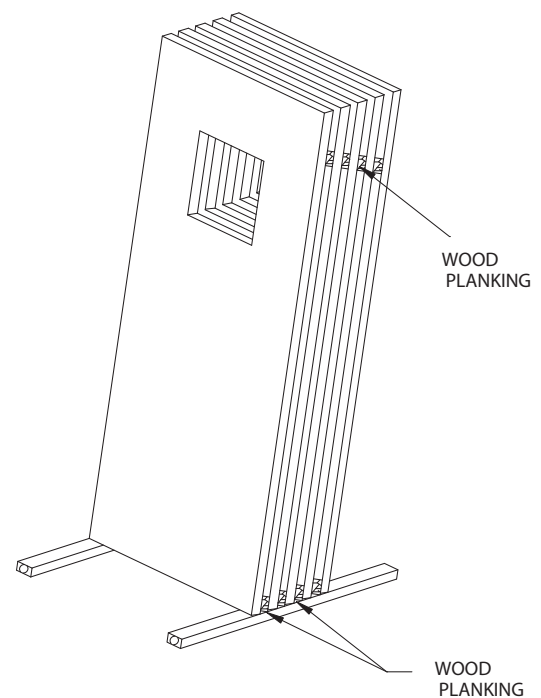


FIGURE 2
DOOR STORAGE

PART 3 - INSTALLATION OF FRAMES

A. GENERAL

Welded door frames are checked at the factory to ensure that they are square and that no jamb twists have occurred during fabrication. Temporary steel spreaders are then attached to the jamb base to minimize misalignment or other damage during handling and shipment. The frames are loaded on the carrier by personnel experienced in frame packing. The spreaders are for shipping and handling purposes only and must be removed before installing the frame.

At no time shall the spreaders be used during installation.

In spite of precautions, frames can and sometimes do arrive at the job site with minor alignment deviations. Minor deviations from true form and alignment can be corrected by the contractor responsible for installation.

When knock-down frames are used the manufacturer's recommended assembly procedures must be followed before installation of the frame can begin. A knock-down frame is easier to pack for shipment and requires less shipping space than a welded frame. Such frames, however, place greater responsibility on the installer, who must assemble them and make sure that, in spite of job conditions, they are square and in true alignment. Frames of this type normally use bendable tabs, screw or bolt fasteners in their assembly, and these may be lost in shipment or at the job site. In general, knockdown frames are not recommended for doors wider than 3 ft. 6 in. (1067 mm) or weighing more than 100 pounds (45 kg).

B. PRIOR TO INSTALLATION

The installer shall perform the following prior to installation:

The area of floor on which the frame is to be installed and the path of the door swing shall be checked for flatness and levelness. Permissible tolerance is +/- 1/16 in. (1.5mm). If the floor exceeds this, it is the general contractor's responsibility to correct the area that is out of tolerance before the frame is installed.

Frame shall be checked for correct size, swing, fire rating and opening number. Remove temporary steel spreaders. Spreader can typically be removed with cold chisel and hammer. With frame standing on concrete, position cold chisel at weld joint of spreader and jamb and strike chisel with hammer.

C. INSTALLATION TOLERANCES

During the setting of the frame, check and correct as necessary for opening width, opening height, squareness, alignment, twist and plumbness. Permissible frame product installation tolerances shall be maintained within the following limits: See Figure 3.

Opening width – Measured horizontally from rabbet to rabbet at top, middle and bottom of frame; +/- 1/32 in. (1mm).

Opening height – Measured vertically between the frame head rabbet and top of floor or bottom of frame minus jamb extension at each jamb and across the head; +/- 1/16 in. (1.5mm).

Squareness – Measured at rabbet on a line from jamb, perpendicular to frame head; not to exceed 1/16 in. (1.5mm).

Alignment – Measured at jambs on a horizontal line parallel to the plane of the face; not to exceed 1/16 in. (1.5mm).

Twist – Measured at opposite face. Corners of jamb on parallel lines, perpendicular to the plane of the door rabbet.

Plumbness – Measured at jambs on a perpendicular line from the head to the floor; not to exceed 1/16 in. (1.5mm).

The tolerances shown provide a reasonable guideline for proper installation of hollow metal frame products. However, it should be noted that the cumulative effect of the installation tolerances at or near their maximum levels could result in sufficient misalignment to prevent the door from functioning properly. Installers should be careful not to create a tolerance buildup. Tolerance buildup occurs when several tolerances are at or near their maximums. Care should be taken to keep each of these tolerances as close to zero as possible.

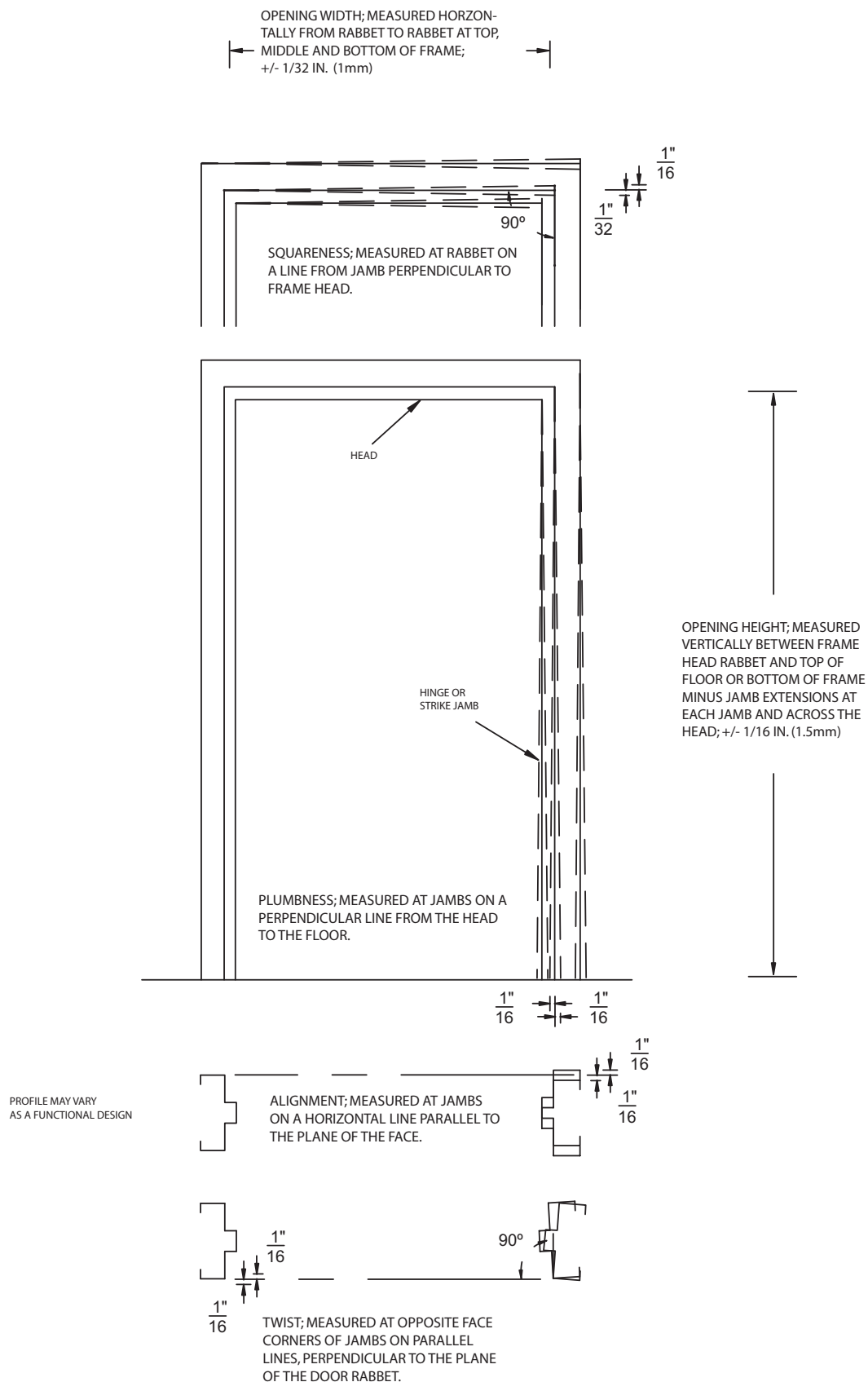
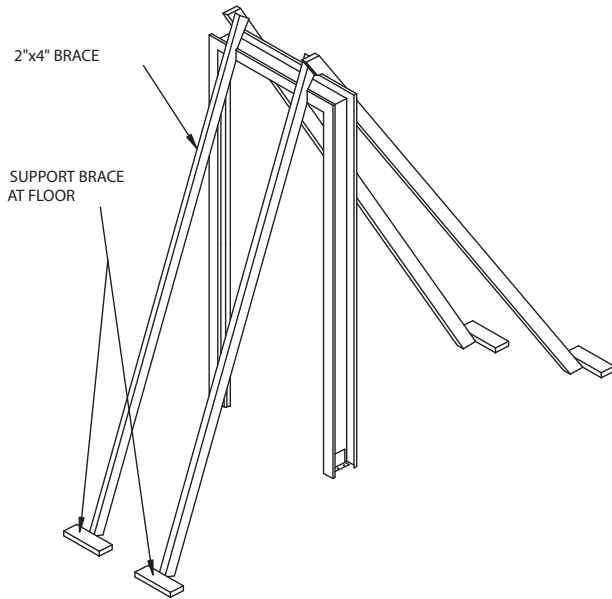


FIGURE 3
FRAME INSTALLATION TOLERANCES

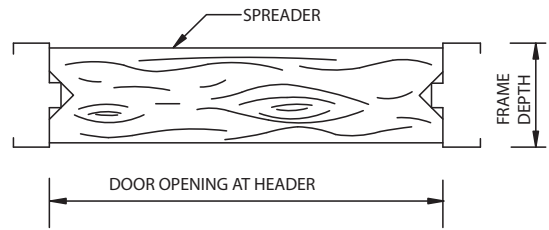
D. TYPICAL INSTALLATION PROCEDURES

Position frame in the correct location. Brace the frame as shown, Figure 4. Do not brace in the direction of intended wall.

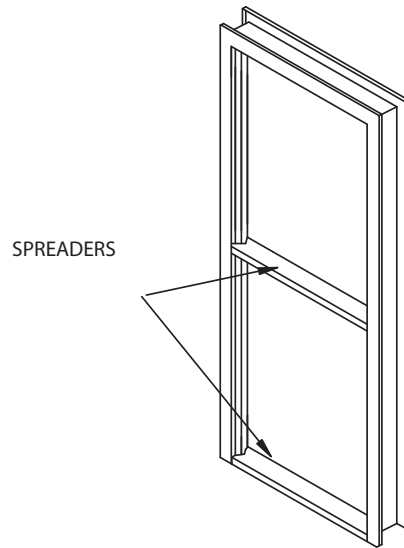


**FIGURE 4
FRAME BRACING**

With frame in position, install the temporary wood spreaders. The wood spreader, Figure 5, must be square and no less than 1 in. (25 mm) thick. Correct length is the door opening width between the jambs at the header. Cut clearance notches for frame stops. Spreader must be nearly as wide as frame depth for proper installation. Install a spreader at the bottom of the frame and a second wood spreader at the mid or strike point to maintain a proper door opening and to prevent bowing of the jambs, Figure 6. Clamp or wire spreaders to frame to hold spreaders in place until the frames are set permanently in the wall.



**FIGURE 5
WOOD SPREADER**



**FIGURE 6
SPREADER LOCATION**

At frames with jamb opening heights greater than 8 ft. (2438mm) or frame face dimensions less than 1-1/2 in. (38mm), install an additional wood spreader. Space wood spreaders at a maximum of 36 in. (914mm) intervals between header and bottom of frame, Figure 7.

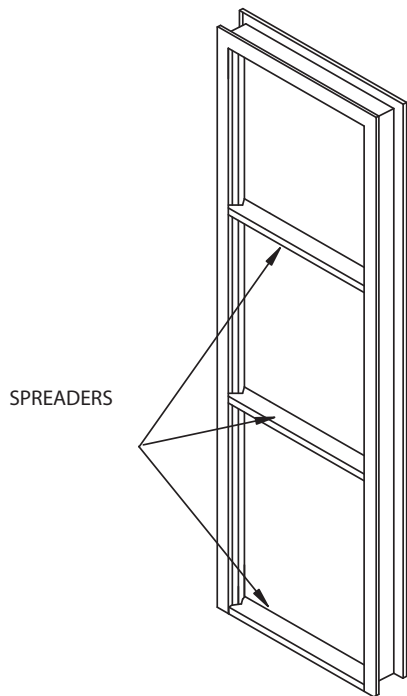


FIGURE 7
SPREADER LOCATION AT
LARGE JAMB OPENING HEIGHTS

At frames with sidelights where the sidelight sill intersects the door jamb near the strike, it is imperative that a wood spreader is located at this location, Figure 8.

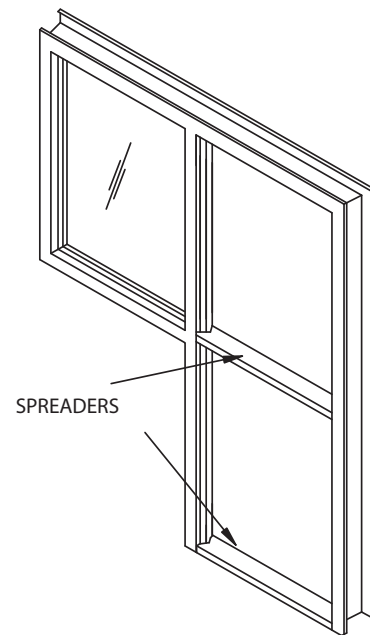
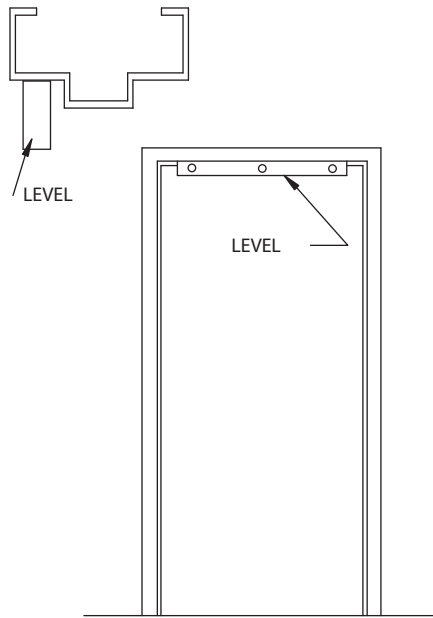


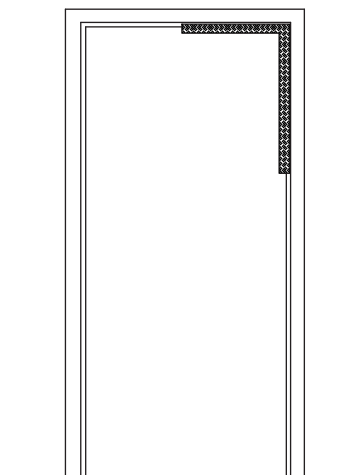
FIGURE 8
SPREADER LOCATION AT SIDELIGHTS

The installation contractor shall have a carpenter level and builder's square. Level the head by positioning the level to the head door rabbet, Figure 9. If necessary, adjust for high spots in floor by shimming under the jamb floor anchor, Figure 13a. Equalize them through an adjustable floor anchor, if specified, Figure 13b. Note, for labeled openings the maximum floor clearance is 3/4 in. (19mm).



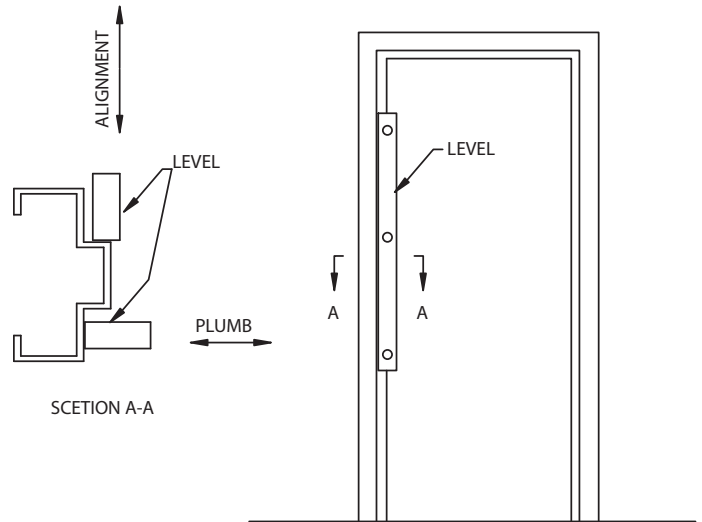
**FIGURE 9
LEVELING THE HEAD**

With builder's square, check frame for squareness. Position square against jamb and head at door rabbet, adjust as required, Figure 10.



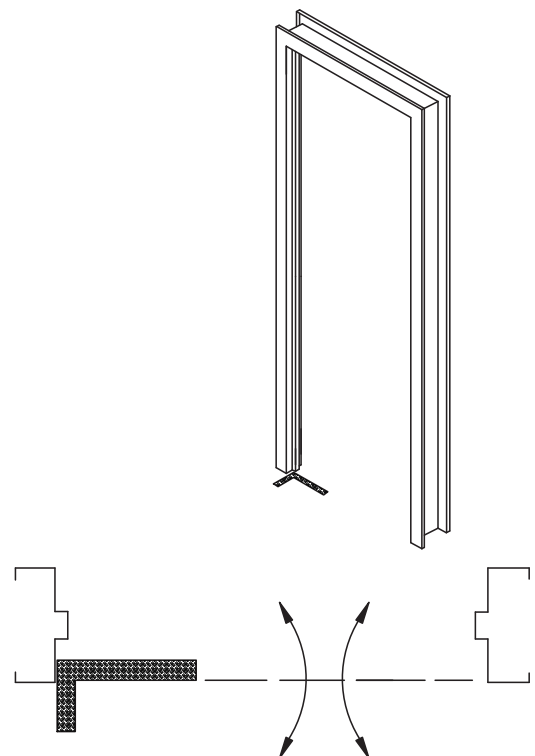
**FIGURE 10
SQUARENESS**

With carpenter level check the frame for plumbness and alignment: For plumbness, position level against both hinge and strike jambs in the rabbet. For alignment; Position level against both hinge and strike jambs on the stop, adjust as required, Figure 11.



**FIGURE 11
PLUMBING AND ALIGNMENT**

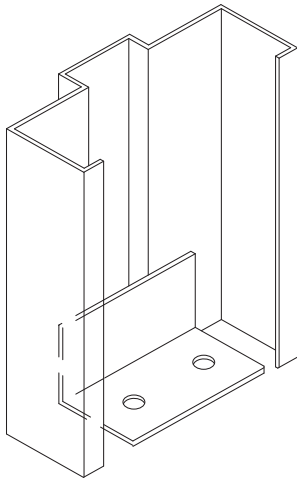
With builder's square, check jambs for twist. Position square against door rabbet and project line perpendicular to the plane of the door rabbet, adjust as required, Figure 12.



Once the installer has ensured that the frame is in the correct position, anchor the jamb to the floor.

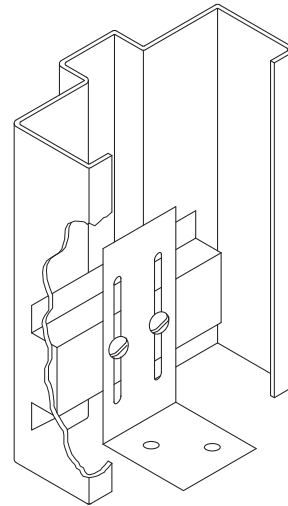
Floor Anchors: The fixed floor anchor is welded to the base of the jamb, typically secured to the floor by mechanical fasteners, providing solid anchorage for the base of each jamb, Figure 13a. Shimming must be used if the floor is not level. This type of anchor is not used in existing masonry, preframed stud walls or slip-on dry-wall frames, but may be provided.

At prepared openings (existing masonry or concrete) an additional expansion anchor reinforcement is provided. With pre-framed stud walls or completed drywall openings an additional base anchor is included. In both instances these anchors are located as close to the bottom of the jamb as is practical.



**FIGURE 13A
FIXED FLOOR ANCHOR**

The adjustable floor anchor is designed for use where there are significant irregularities from level, an intended slope in the floor, or where toppings such as terrazzo are used, Figure 13b. Like the standard anchor, it cannot be used in prepared openings or slip-on knock-down dry-wall frames.



**FIGURE 13B
ADJUSTABLE FLOOR ANCHOR**

E. WALL CONSTRUCTIONS AND APPROPRIATE ANCHORS

Proper frame anchoring is vitally important to the proper performance of the door opening. A variety of jamb wall anchors are available to suit the various types of wall construction. Anchors shown on the following pages are but a few of the variety of anchoring methods available.

ANCHOR SPACING

Anchor Spacing: CSDMA specifications require that the number of wall anchors used on each jamb, in all cases, be as follows:

- A. Openings in Masonry Walls. Frames with expansion bolt anchors; Anchors spaced a maximum of 6 in. (152 mm) from the top and bottom, with intermediate spacing at a maximum of 26 in. (660 mm) o/c. See Table 1.
- B. Openings in stud partitions with steel or wood stud anchors: Near hinges and directly opposite on strike jamb. See Table 1.
- C. Anchor quantity for slip-on drywall frames shall be as per manufacturer's standards and label requirements.
- D. Wall anchors locations and quantity are in addition to floor anchors, when required.

Table 1

Frame Anchor Spacing

Frame Height	Masonry Walls	Stud Partitions
> 0 in. < 60 in. (1524 mm)	2	3
> 60 in. < 90 in. (2286 mm)	3	4
> 90 in. < 96 in. (2438 mm)	4	5
> 96 in.	4 + 1 per 24 in. (610 mm)	5 + 1 per 24 in.

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METAL DOOR LTD

Installation Troubleshooting Guide for Standard Steel Doors and Frames

Metal Door Ltd. does not condone or encourage repair methods which would adversely affect product performance or violate and/or void product warranties.

The user of this document assumes all responsibility associated with but not limited to product performance and violation of product warranties for any product associated with the installation and suggested repair methods in this document.

PURPOSE

The intent of this document is to cover field installation problems most commonly experienced with standard steel door and frame installations. It should be understood that most problems encountered are because of inappropriate application of the products and/or improper installation.

The suggested method of repairs requires only basic hand tools and relatively little time. It should be understood that more complex problems or compound problems could exist which warrant extensive field repairs and modification to products. These types of field installation problems are not intended to be covered in this document and should not be made without first consulting the manufacturer.

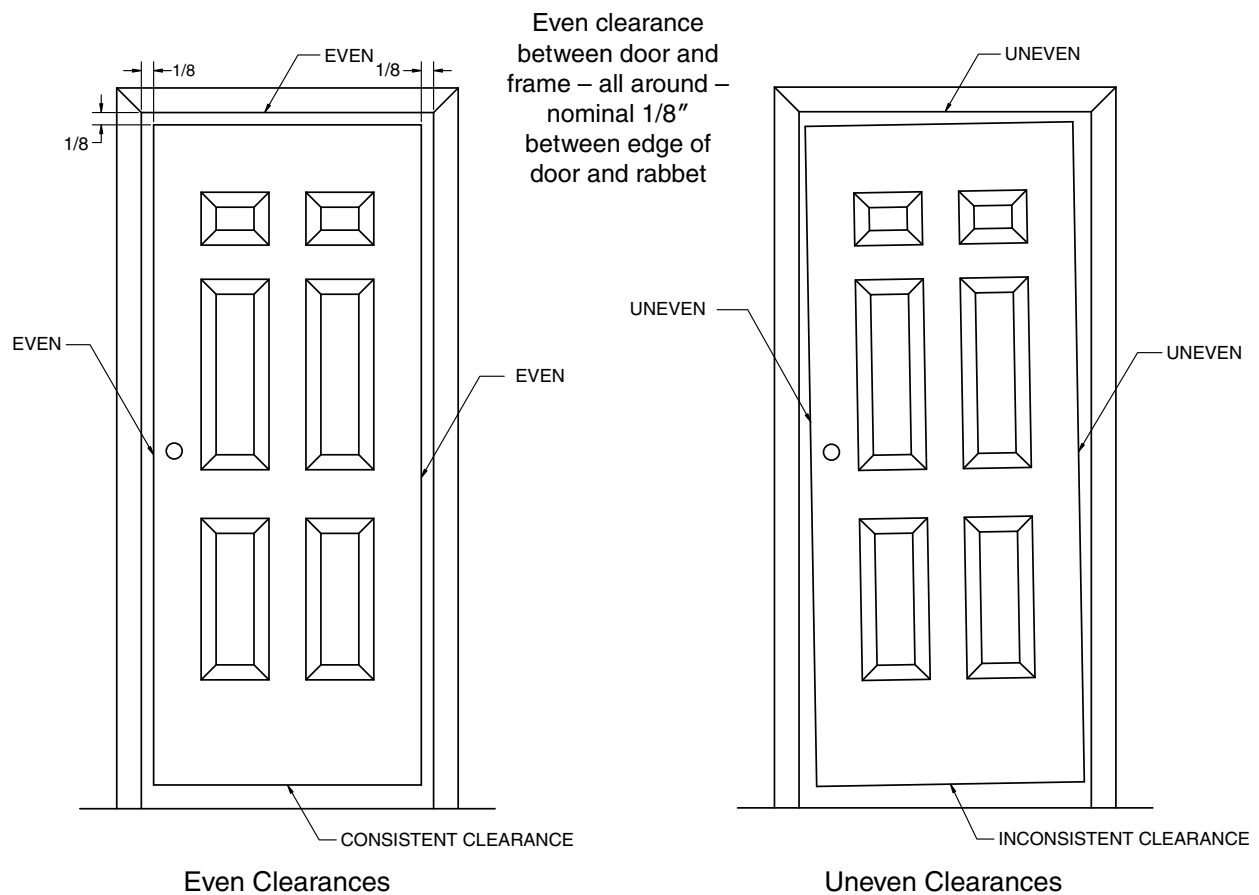
Modifications made to fire-rated-labeled doors and frames shall be made in compliance with NFPA 80.

Installation Troubleshooting Guide for Standard Steel Doors and Frames

IMPROPER DOOR / FRAME CLEARANCE

To ensure proper door and frame function, the clearance between the door and frame must be properly maintained. All standard steel door and frame manufacturers closely hold tolerances that result in a nominal clearance between the door and frame of 1/8 inch. If this clearance is not maintained, interference and hardware misalignment may occur.

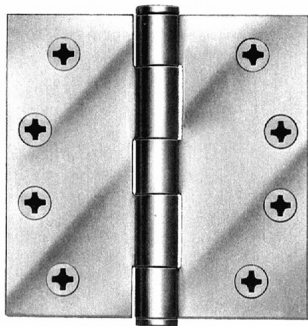
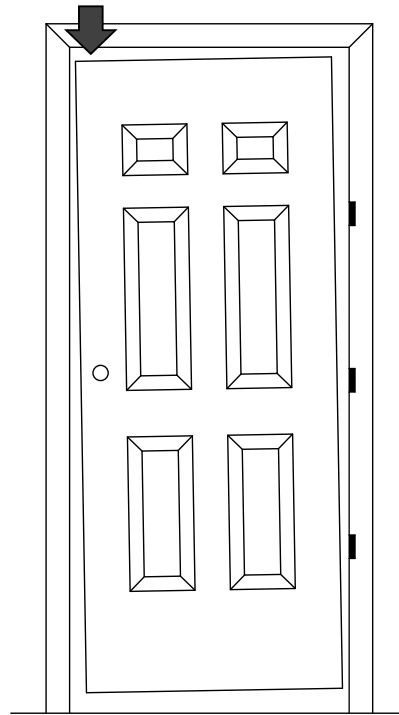
Proper installation is extremely important in establishing clearances and preventing a multitude of potential problems.



IMPROPER DOOR / FRAME CLEARANCE *(continued)*

Is door sagging?

Uneven clearances frequently result from improper installation of frames and frame anchors. The condition, characterized by lock edge clearances narrow at the top and wide at the bottom, is called “door sag”. The following suggested adjustments may correct the condition.



Are hinges loose?

If hinge screw does not remain tight, the screw can be held in place by the use of a “locktite” type product which prevents the screw from loosening. Additionally, “Nyloc” type fasteners can be used to replace the normal machine screws.

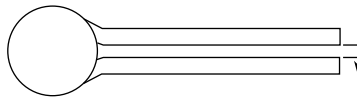
Are hinges worn?

If excessive wear has occurred on hinge knuckles, the door will not be held tightly. Replace hinges.

Are hinges properly swaged?

The hinge manufacturer’s specifications should be checked to determine the proper hinge swage.

Swaging is a slight offset of the hinge leaf at the barrel which permits the leaves to come together.

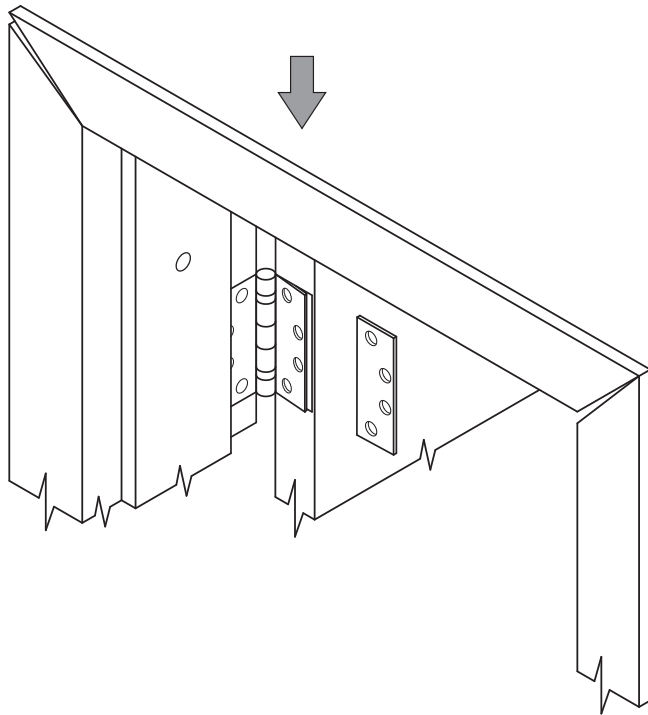
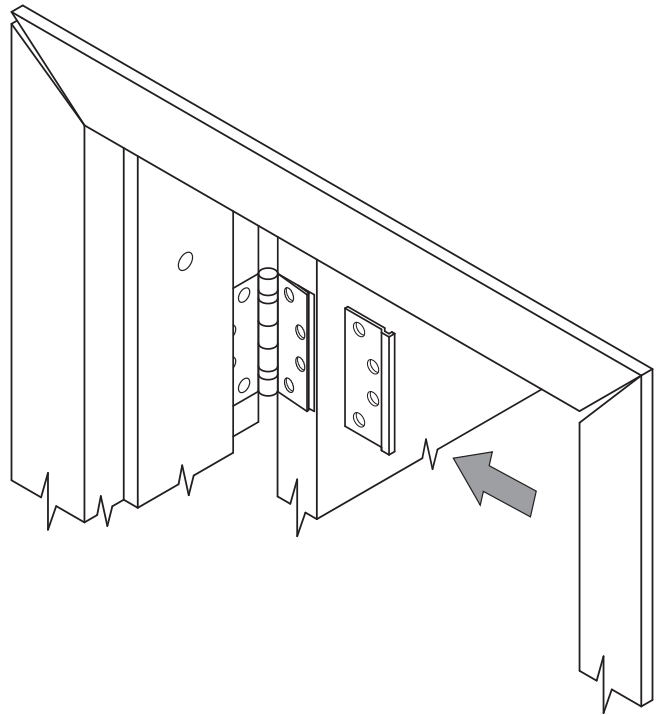


Standard swaging of standard-weight and heavy-weight full mortise hinges, when closed to the parallel position, provides a $\frac{1}{16}$ ” clearance between leaves.

IMPROPER DOOR / FRAME CLEARANCE *(continued)*

Remove top hinge filler for non-handed doors

The improper clearance condition may be improved by removing the top hinge leaf attached to the door and taking out the filler (hinge filler) from behind the leaf. Once completed, reattach hinge leaf to door. A hinge filler is used on all non-handed doors.



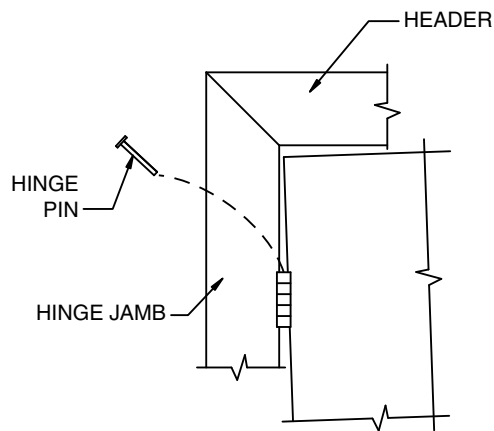
Add shims to bottom and/or middle hinge

On handed or non-handed doors, the lower half of the door can be shifted upward and toward the strike jamb. Remove the screws which attach the middle and/or bottom hinge(s) to the door. Insert shims between the hinge leaf and door. Reattach hinge leaf to door.

Standard-weight to heavy-weight hinge modifications

Although styles and types vary, many manufacturers offer built-in features that allow hinge pockets on doors and frames to be modified from standard-weight to heavy-weight. Examples of the different methods are: grinding or flattening down raised embossments in shim plates or reinforcements; removing or breaking off hinge filler shim plates; adjusting set screws; and removing wire shims. These options all involve removing material from the pocket to allow for heavy-weight depth to be achieved. As always, check with individual manufacturers to determine the type of modifications offered.

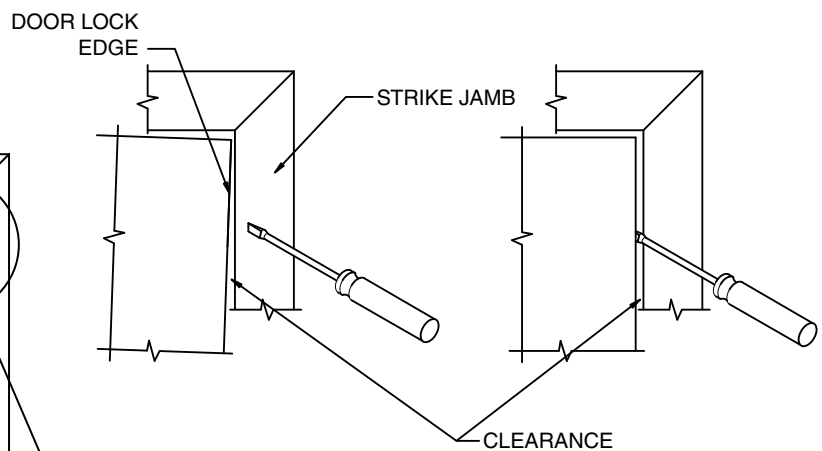
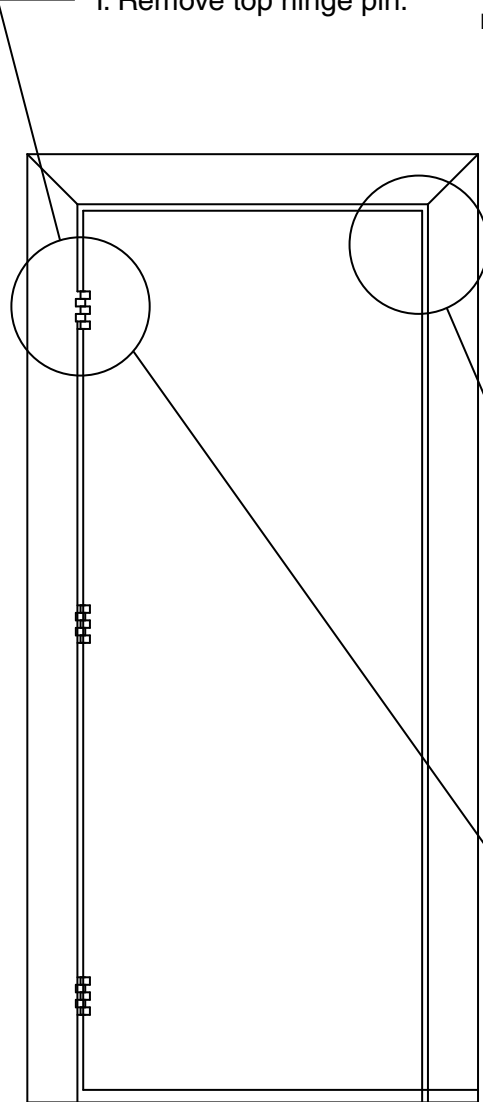
IMPROPER DOOR / FRAME CLEARANCE (continued)



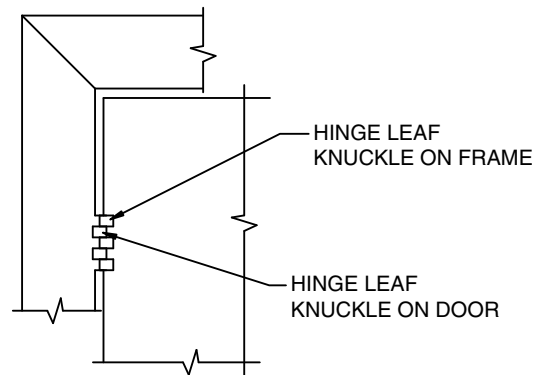
Reswaging hinges

The following example shows how a hinge leaf can be reswaged to correct minor improper door/frame clearances. This particular method allows the reswaging to be accomplished while the door remains in the opening and the hinge leaves remain on the door and frame. The example shows a top hinge reswaged to correct a sag-type condition. However, any of the hinges can be reswaged in this manner to compensate for conditions opposite to that of a sag condition.

1. Remove top hinge pin.



2. Use a screwdriver to set the clearance between the lock edge of the door and strike jamb.

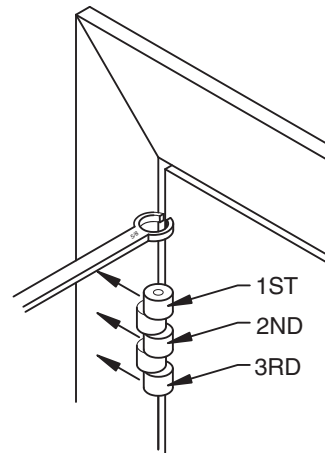


3. Hinge knuckle will now be misaligned.

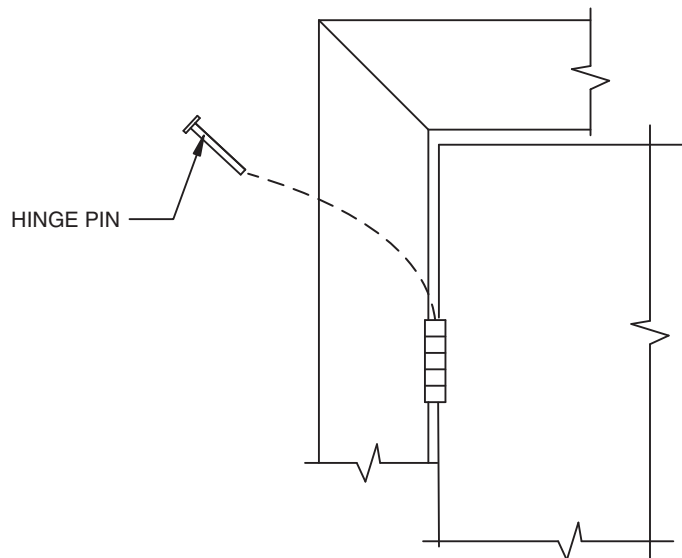
IMPROPER DOOR / FRAME CLEARANCE *(continued)*

Reswaging hinges *(continued)*

As shown, the hinge leaf knuckles on the door move to a new location once the clearance at the strike jamb is set. The hinge leaf knuckles on the frame must now be reswaged (bent) to align with the new location of the hinge leaf knuckles on the door. The simplest way to do this is to use a $\frac{5}{8}$ " – $1\frac{1}{16}$ " tube wrench (a crescent wrench will do as a second choice). Simply slip the tube wrench down around the 1st knuckle and align it with the new location. Once the first knuckle has been aligned, the wrench will now slip down to the 2nd knuckle for alignment. Repeat the process on the third knuckle.



4. Reswage and align hinge knuckles.



NOTE: As always, caution should be exercised when the hinge leaves are not removed. **Excessive** pressure could damage doors/frames. As an alternate reswaging method, the hinge can be removed and clamped in a vise for reswaging.

5. Re-insert the hinge pin and remove the screwdriver.

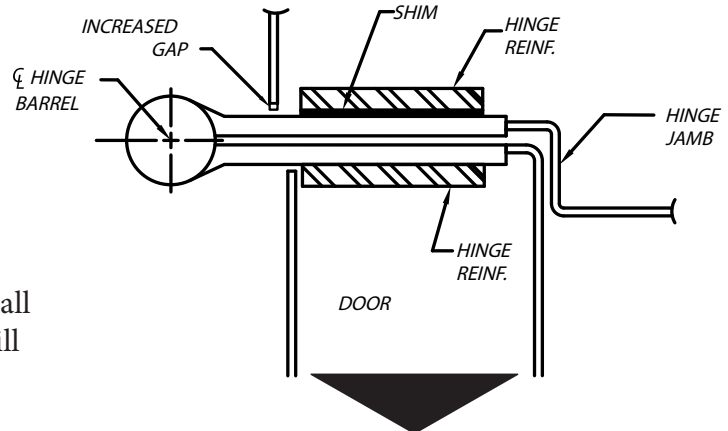
Is the door binding?

Frames which are out of plumb will likely cause improper operation of locksets and binding of bolts in the strike. Check carefully the installation of the frame prior to making hardware adjustments.

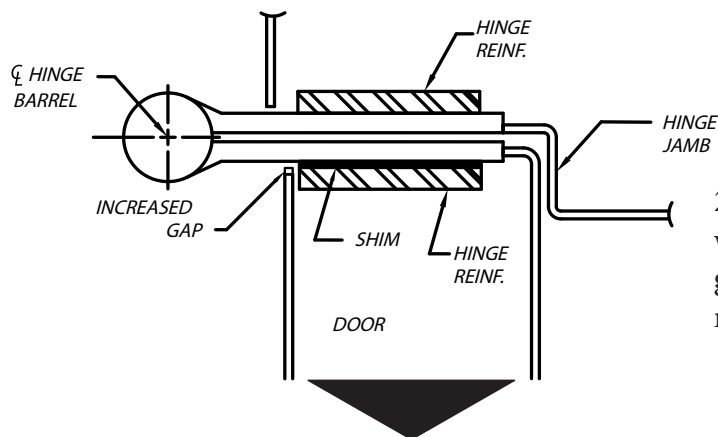
DOOR BIND AGAINST RABBET

Normally, door bind is found between the door and rabbet. There are several ways of shimming which will move the door in different directions. The following guidelines should be used in shim applications.

2.1 This method will move the door toward the Strike jamb. It will leave a small gap between Hinge and Frame face. It will not prevent or help hinge bind.

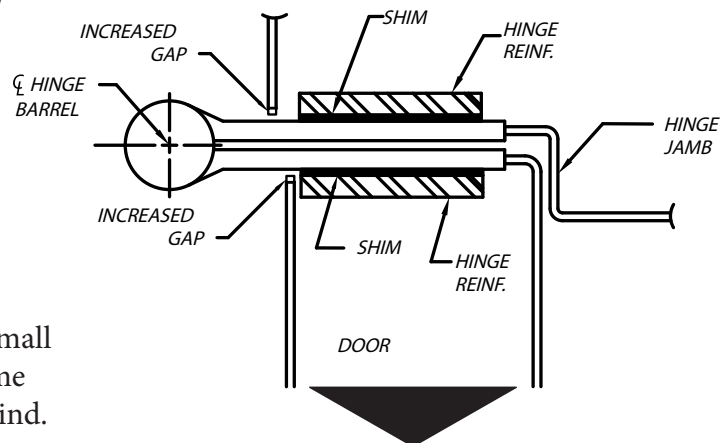


*Door - Frame - Hinge
Full Shim - Frame*



*Door - Frame - Hinge
Full Shim - Door*

2.2 This method will move the door toward the Strike jamb. It will leave a small gap between Hinge and Door face. It will not prevent or help hinge bind.



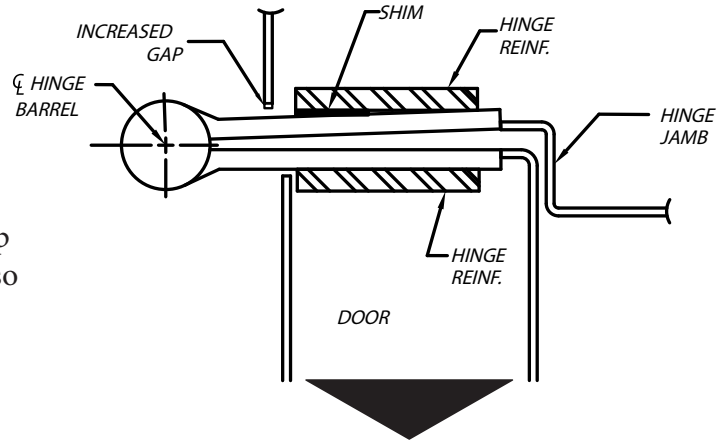
*Door - Frame - Hinge
Full Shims - Door & Frame*

2.3 This method will move the door toward the Strike jamb. It will leave a small gap between Hinge and Door and Frame face. It will not prevent or help hinge bind.

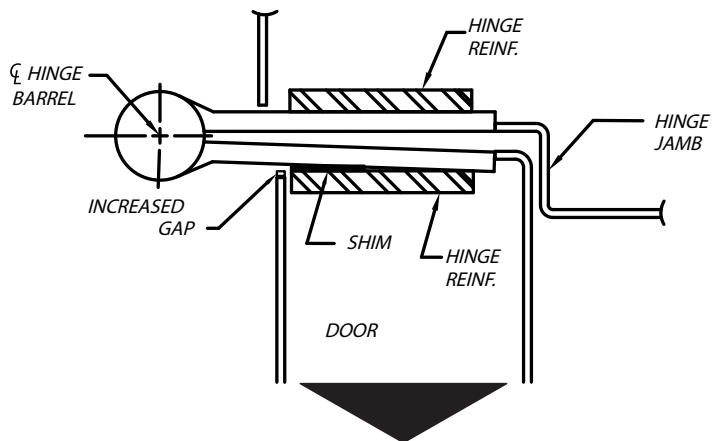
HINGE BIND AGAINST STOPS

Hinge bind against the frame stops is possible field condition. There are several ways of shimming which will move the door in the desired direction. The following guideline should be used in shim application.

3.1 This method will move door toward Strike jamb. It will leave a small gap between Hinge and frame face. It will also help if hinge bind is occurring.

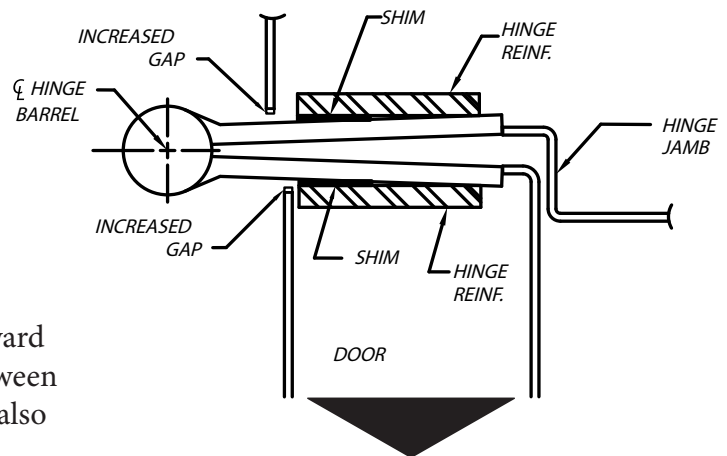


*Door - Frame - Hinge
Half Shim - Frame*



*Door - Frame - Hinge
Half Shim - Door*

3.2 This method will move door toward Strike jamb. It will leave a small gap between Hinge and Door face. It will also help if hinge bind is occurring.

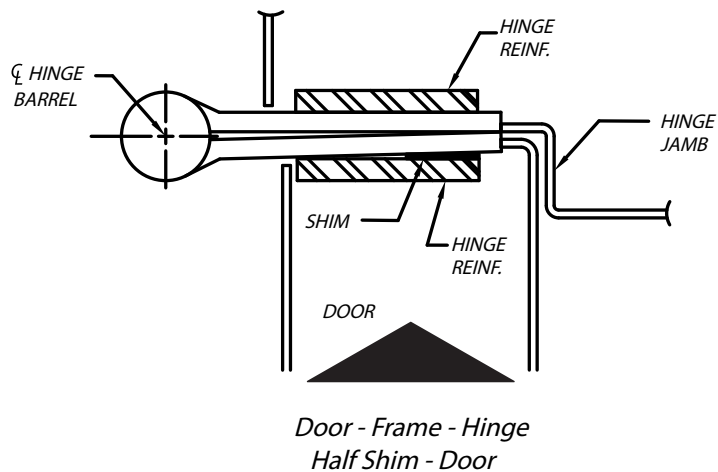
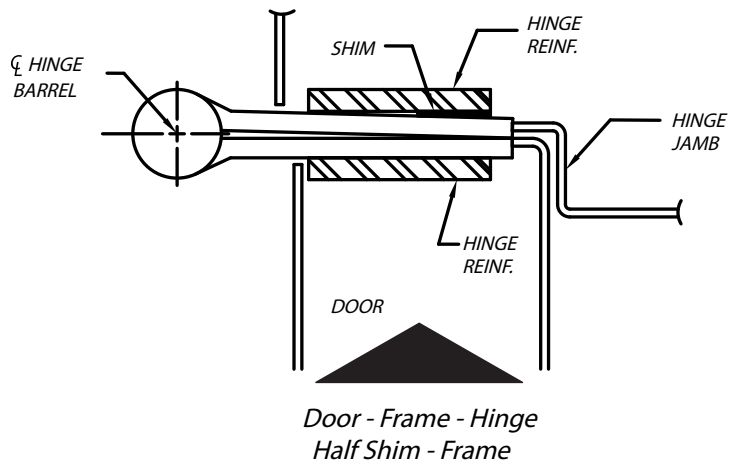


*Door - Frame - Hinge
Half Shims - Door & Frame*

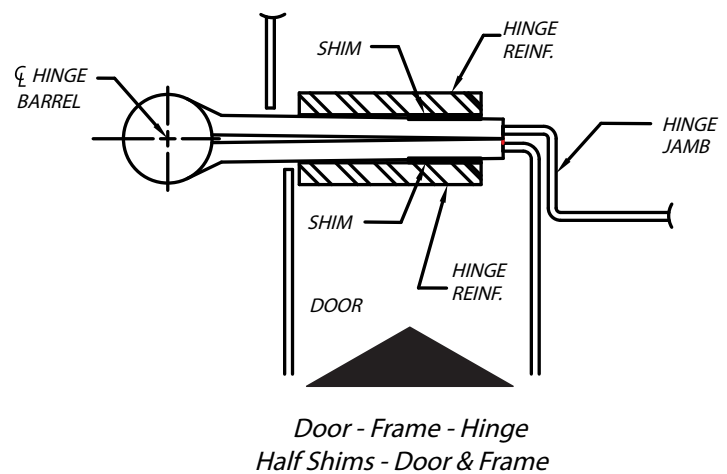
3.3 This method will move door toward Strike jamb. It will leave a small gap between Hinge and Door and Frame face. It will also help if hinge bind is occurring.

HINGE BIND AGAINST STOPS (continued)

4.1 This method will pull door toward Hinge jamb. It may create hinge bind.



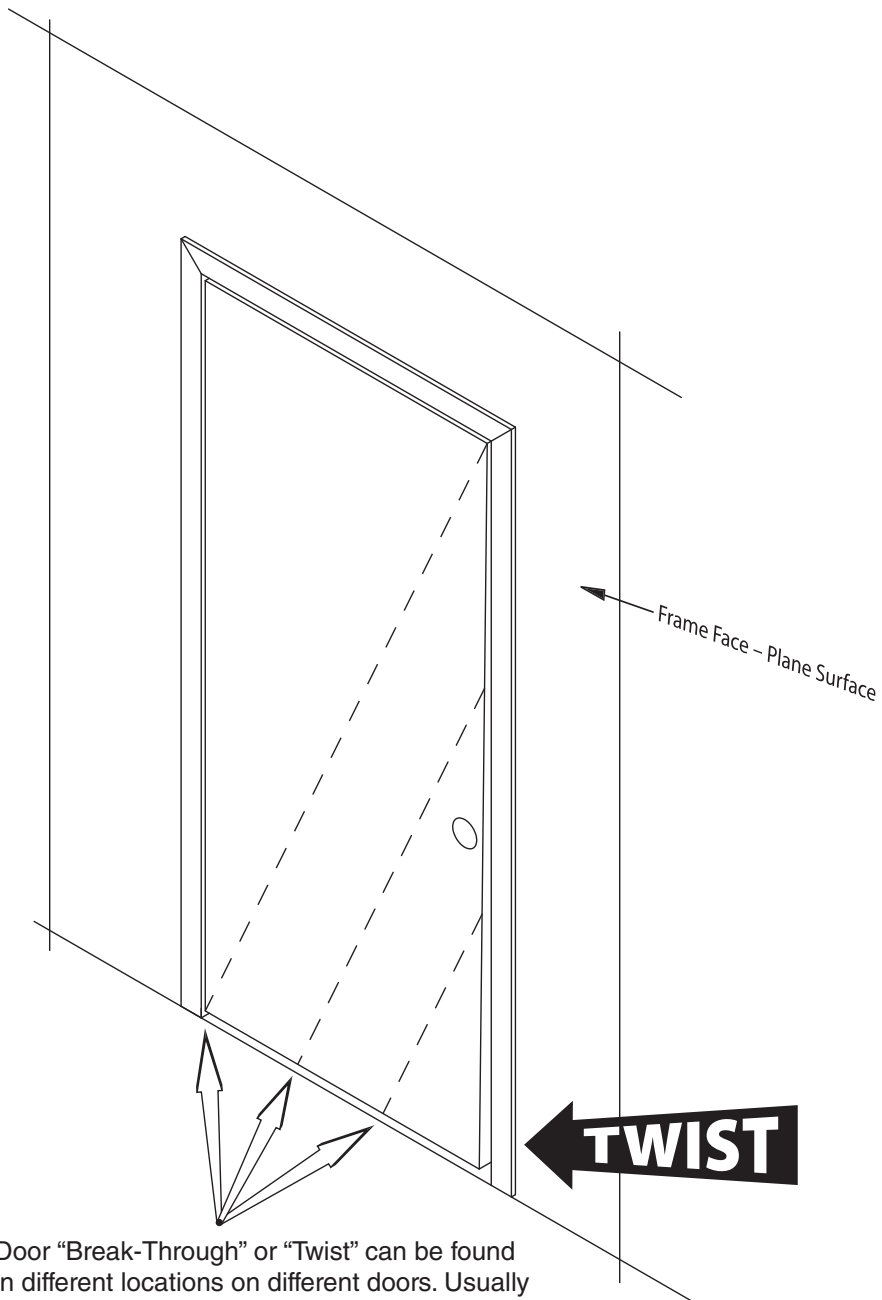
4.2 This method will pull door toward Hinge jamb. It may create hinge bind.



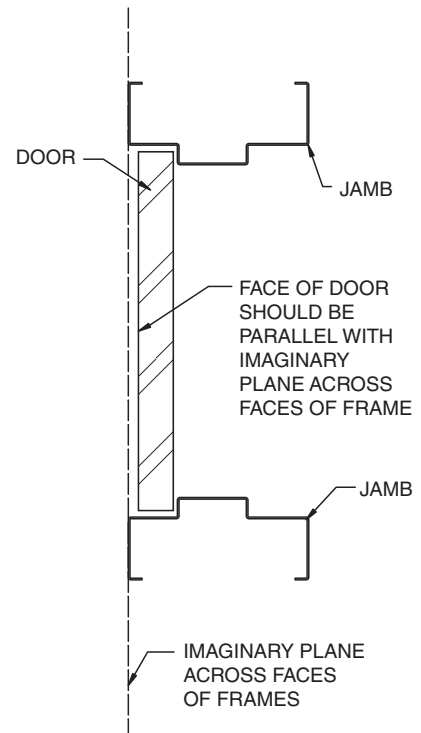
4.3 This method will pull door toward Hinge jamb. It may create hinge bind.

TWISTED DOOR

Normal installation results in the plane of the door face being parallel with the plane of the frame face. If the frame is square and plumb, all face surfaces of the frame will be in the same plane. A twisted door will “break through” the frame’s face plane surface. **BE SURE** the frame is square and plumb. If it is not, the problem is probably with the frame installation and **NOT** the door.

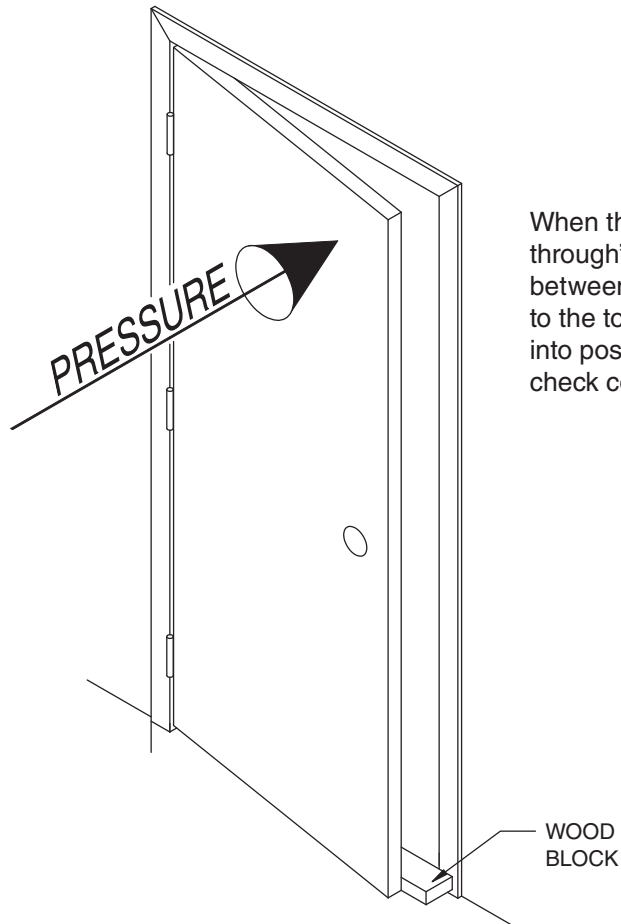


Door “Break-Through” or “Twist” can be found in different locations on different doors. Usually it is found diagonally between opposite lower and upper corners.



TWISTED DOOR ADJUSTMENTS

When required, it is possible to “spring” the door back to (or much closer to) its ideal position of being parallel with the plane across the faces of the frame. This can usually be done with the door remaining in the frame. A piece of wood blocking must be placed between the door and frame. Pressure is then applied at the twisted area to “spring” the door. However, caution should be exercised on drywall installations since the frame could possibly work loose from the wall, particularly with slip-on drywall type frames.

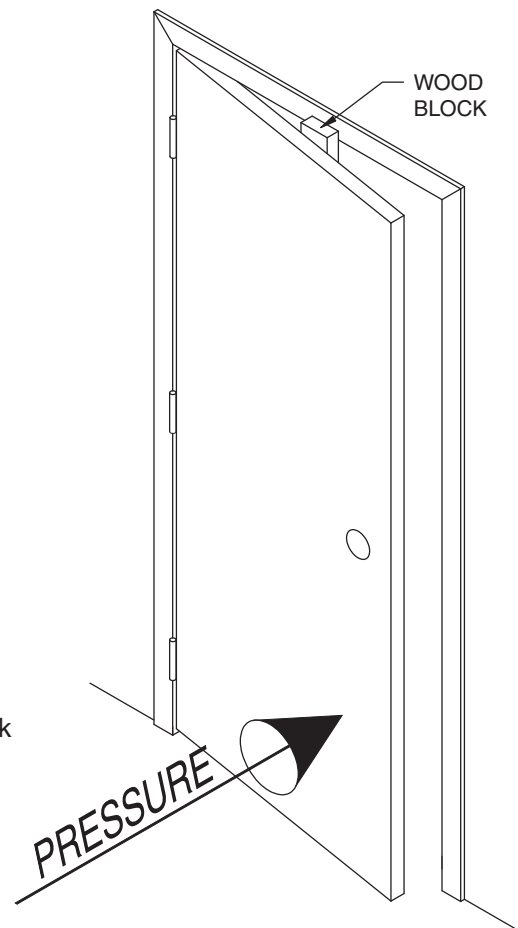


Twisted at Top

When the top lock area of a door is “breaking-through” the plane, place a wood block on floor, between door and frame as shown. Apply pressure to the top lock area as shown to “spring” door back into position. Remove wood block, close door and check condition. Repeat if necessary.

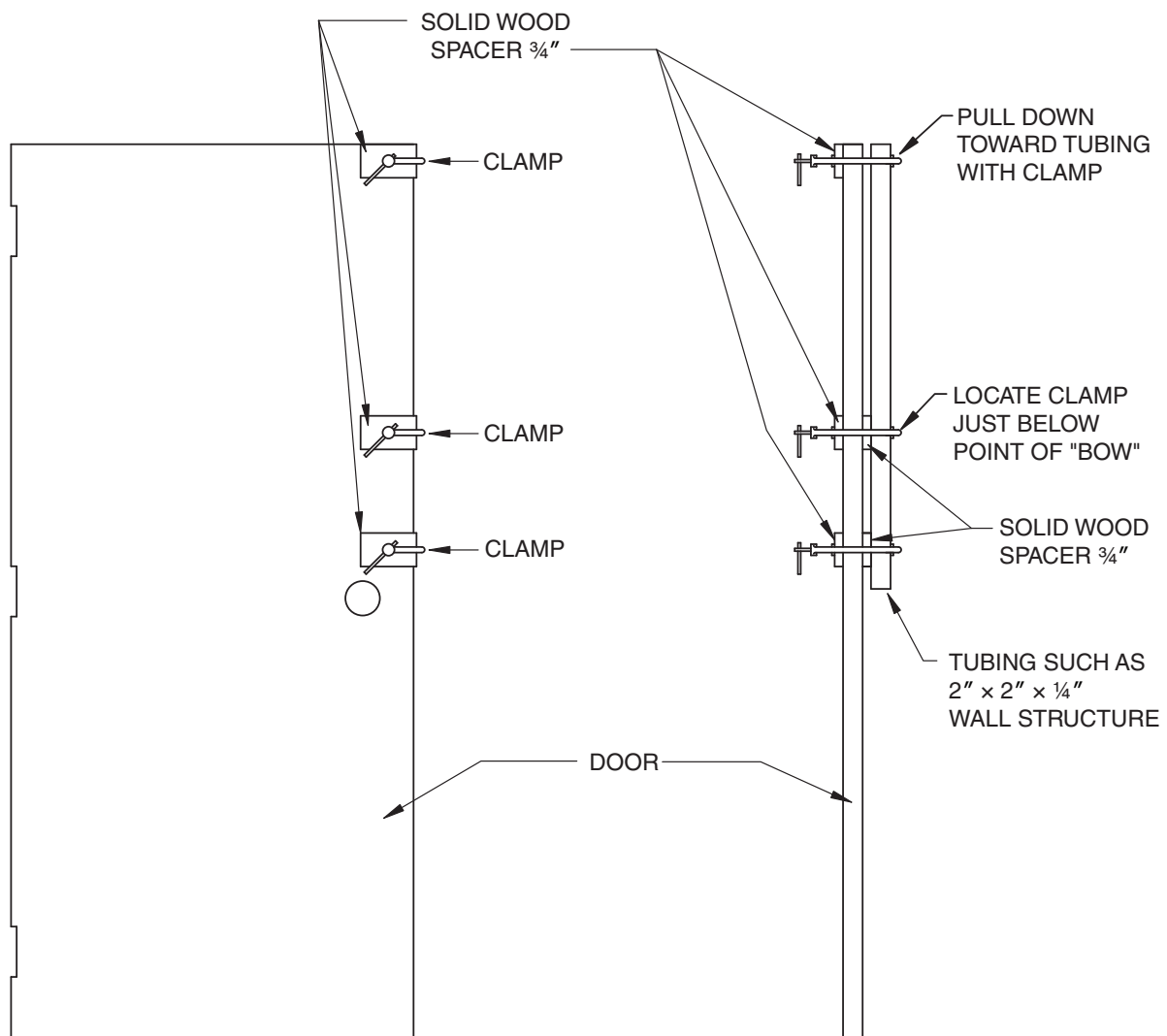
Twisted at Bottom

When the bottom lock area of door is “breaking-through” the plane, place a wood block between frame head and door as shown. Apply pressure to the bottom lock area as shown to “spring” door back into position. Remove wood, close door and check condition. Repeat if necessary.



TWISTED DOOR ADJUSTMENTS *(continued)*

An alternate method can also be used which will allow the door to remain in the opening. This method might be appropriate in drywall installations as previously mentioned. Although the example shown deflects the top half of the door, this method could be used on the bottom half of the door as well.

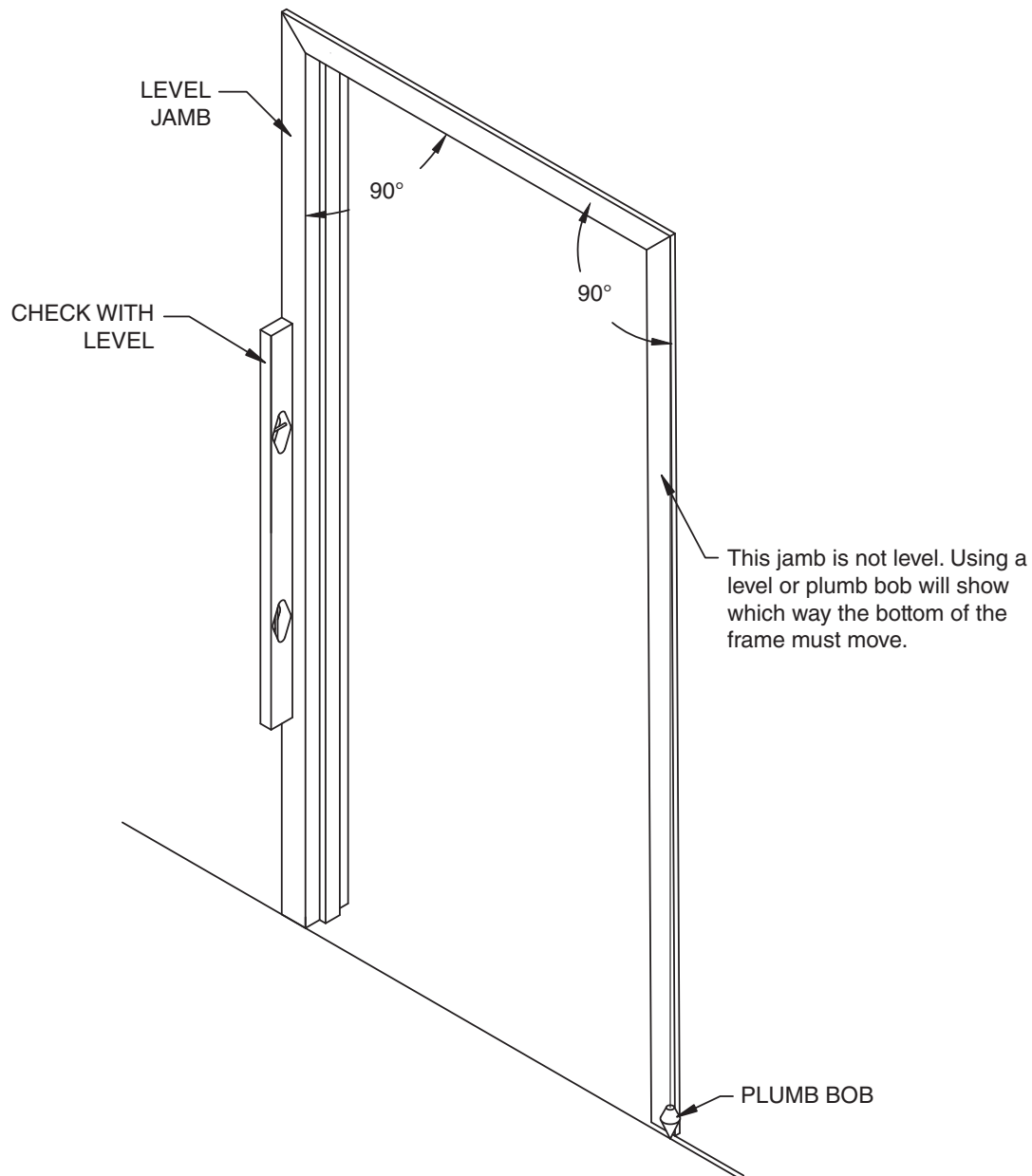


TWISTED FRAME

Wall conditions and anchoring methods can cause the frame to be “twisted” in the opening. The two jamb (hinge and strike) faces are not in the same plane as discussed in the twisted door section. This can be checked by using a level and/or plumb bob. Frames do not have adjustments when it comes to “twist.”

The “twist” condition is generally caused by the wall conditions. The frame jamb faces are prevented from being set in the same plane because the walls are out of plane to begin with.

The options available to correct this problem depend, to a large extent, on the wall construction.

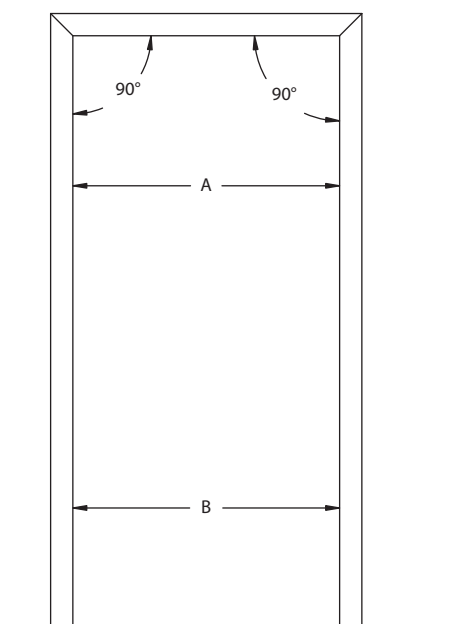


FRAME SET OUT OF SQUARE

Proper frame installation cannot be overemphasized. In the majority of wall constructions, ("slip-on" drywall type frames being the major exception) "quick" or "easy" field fixes are limited. Improperly set frames that are drastically out of square will cause severe door and hardware problems. These types of situations would require extensive field repair and may result in having to remove frames from walls.

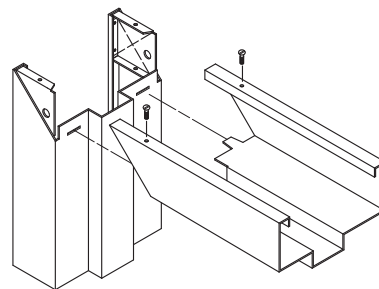
If the "out of square" condition is slight, some adjustment can be made by shimming and/or adjusting anchors, particularly if the frame being worked on is a slip-on drywall type.

The most versatile frame to work with in correcting these types of conditions is the "slip-on" drywall frame. The manufacturers' instructions should be referenced, but generally most "slip-on" drywall frames have a sill anchor at the bottom of the jamb and a compression anchor between the top hinge and the header. Also, most jambs have corner clips that have a hole to accept a screw. The corner clip and hole align with a hole in the header. The adjustment of slip-on drywall frames can be made using these anchors or corner clips.

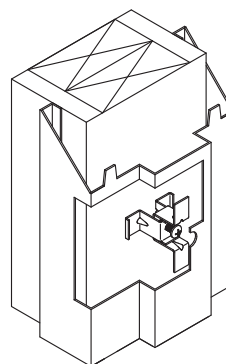


PROPERLY SET FRAME

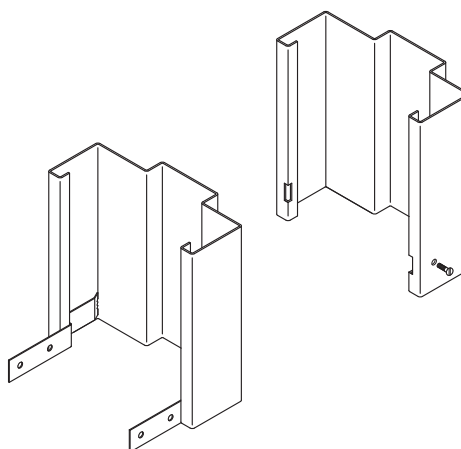
All corners are 90°. The dimension taken at Point "A" is equal to the dimension checked at Point "B." Both hinge and strike jambs are plumb and level.



**Corner Clip
Header to jamb connection**



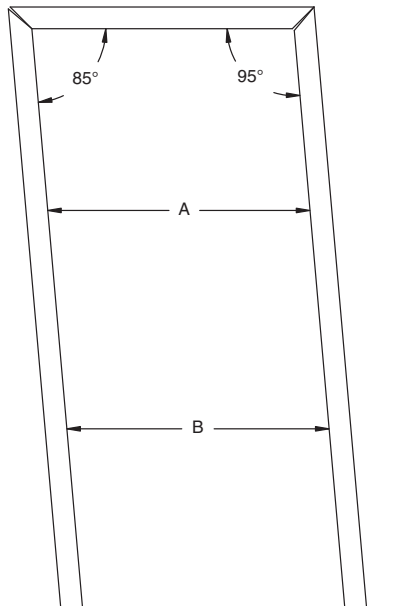
**Compression Anchor
(Cut-away view)**



**Sill Anchors
Snap-in type or screw type**

FRAME SET OUT OF SQUARE (continued)

The following examples illustrate how a frame can be set out of square.



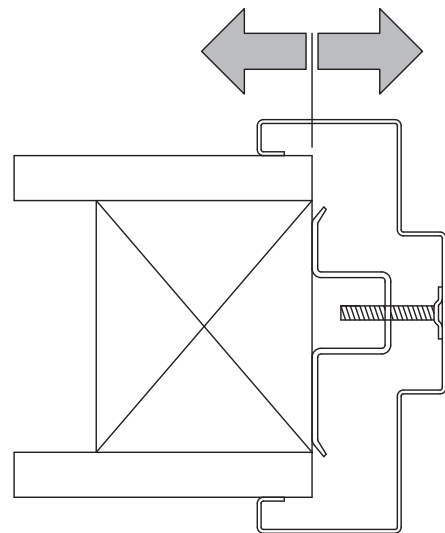
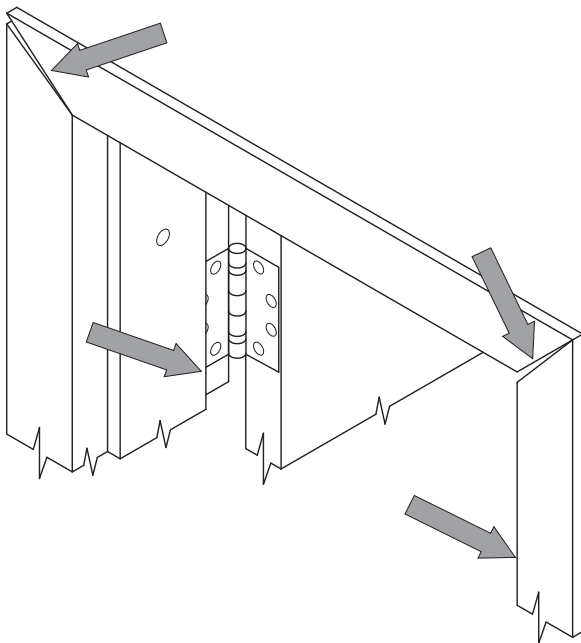
IMPROPERLY SET FRAME (A)

One corner is greater than 90° and one corner is less than 90°. The miters do not properly align with one another and “gaps” are created in the miters which are opposites of one another. The dimension taken at point “A” would remain the same when checked at point “B” if the 90° corners are out of square an equal number of degrees.

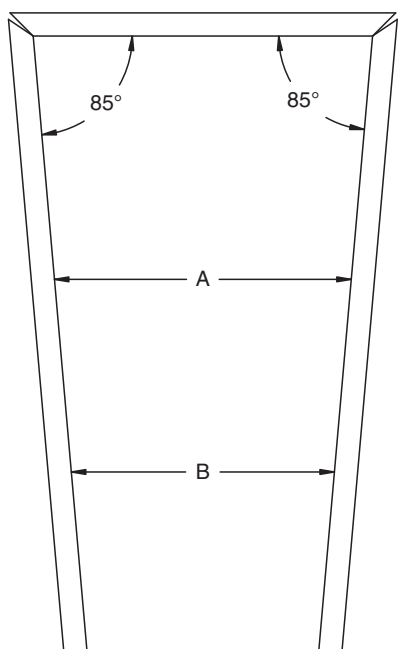
This condition can be corrected by adjusting the compression anchors in and/or out of each jamb as required. Loosen the corner clip screws if the frame is so equipped before making adjustments; retighten when finished. The compression anchors must be adjusted (turned) in opposite directions to allow the jambs to move.

For the illustration shown, the strike jamb compression anchor screw must be turned counterclockwise to retract the compression anchor and give room for the frame to be adjusted back into position using the hinge jamb compression anchor. The hinge jamb compression anchor screw is then turned clockwise to advance the anchor and push the frame towards the room made at the

strike jamb. The frame should be brought into square with the header and the hinge jamb compression anchor. Once this is done, the strike jamb compression anchor screw can be turned clockwise, advancing the anchor snug against the stud. The corner slip screws should then be retightened.



FRAME SET OUT OF SQUARE (continued)



Jambs Toed In at Sill

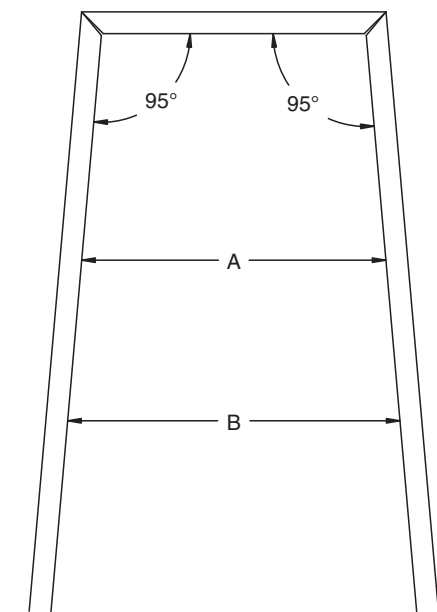
To correct this condition, the sill anchors must be loosened to allow the bottom of each jamb to be moved in and/or out. In the case of the snap-in type sill anchor, the baseboard trim might have to be removed to gain access to the anchors. This type of anchor is generally used because it can be hidden by covering it with the baseboard trim.

If the jambs are toed in at the sills, the compression anchors might have to be retracted slightly so the sill (base) of the jambs can be pushed out. The jambs should be plumbed, squared with the header and leveled as they are pushed out into their proper position. The sill anchors should be reset, the compression anchors adjusted and retightened. The baseboard trim should be reattached to the wall if it had been removed.

A similar procedure should be used if the jambs are toed out at the sills. The sill (base) of the jambs should be pushed in towards the opening. The jambs should be plumbed, squared with the header and leveled as they are pushed into their proper position. The sill anchors should be reset, the compression anchors adjusted (extended) and retightened. The baseboard trim should be reattached to the wall if it had been removed.

IMPROPERLY SET FRAME (B)

The corners are both greater than or less than 90°. The miters do not properly align with one another. When the corner is less than 90° the miter will have a gap which increases as it goes towards the return. When the corner is greater than 90° the miter will have a gap, which increases as it goes towards the rabbet. Dimensions taken at point "A" and "B" will not be equal. This condition will always result in dimension "B" being either greater than or less than the nominal opening dimension which would be obtained if checked between jambs along the header.



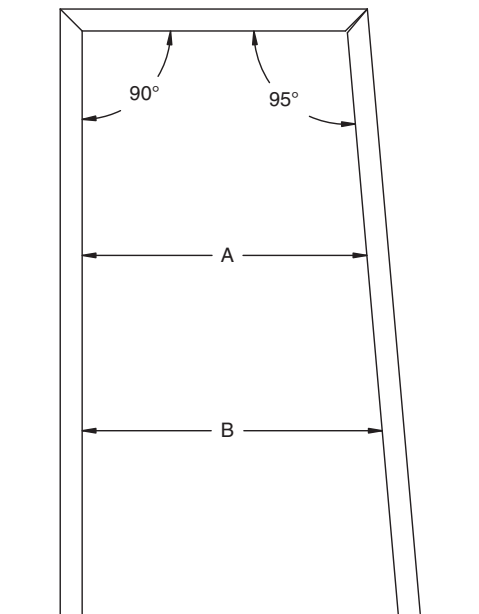
Jambs Toed Out at Sill

FRAME SET OUT OF SQUARE (continued)

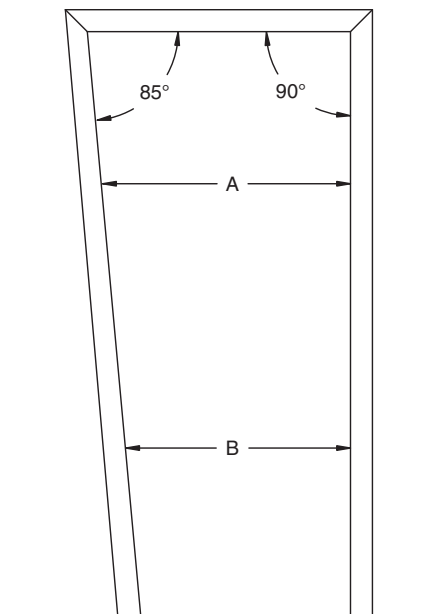
IMPROPERLY SET FRAME (C)

One corner is properly set at 90° while the other corner is greater or less than 90° . The dimension taken at point "A" will constantly change when checked at various points going towards point "B." The corner miter will not properly align between the header and jamb which is not set at 90° .

The jamb which is not 90° to the header must be moved. To correct this condition, the sill anchor must be loosened to allow the bottom of the jamb to be moved in or out as required. In the case of the snap-in type sill anchor, the baseboard trim may have to be removed to gain access to the anchor. This type of anchor is generally used because it can be hidden by covering it with the baseboard trim.



One Jamb Toed Out



One Jamb Toed In

If the jamb is toed in at the sill, the compression anchor may have to be retracted slightly so the sill (base) of the jamb can be pushed out. The jamb should be plumbed, squared with the header and leveled as it is being pushed out into its proper position. The sill anchor should be reset, the compression anchor adjusted and retightened. The baseboard trim should be reattached to the wall if it had been removed.

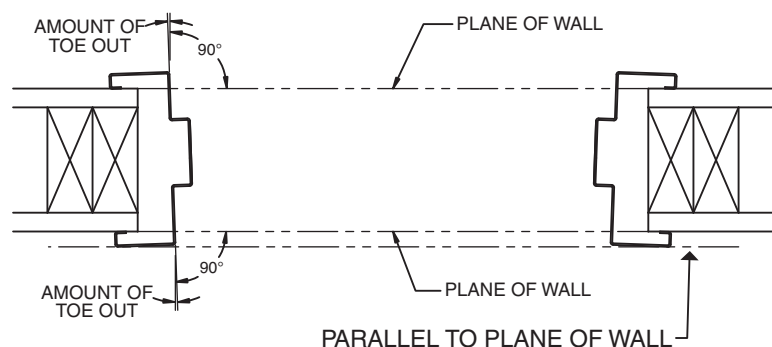
A similar procedure should be used when the jamb is toed out at the sill. The sill (base) of the jamb should be pushed in towards the opening. The jamb should be plumbed, squared with the header and leveled as it is pushed into its proper position. The sill anchor should be reset, the compression anchor adjusted (fastened) and retightened. The baseboard trim should be reattached to the wall if it had been removed.

FRAME WITH RABBETS TOED IN OR OUT

The importance of proper initial frame installation is evident in this condition. In the majority of wall constructions, except for the “slip-on” drywall type frames, “quick,” “easy” field fixes are limited. Improperly set frames which have the rabbets drastically toed in or out would require extensive field repair and in most cases require that the wall be entered to gain access to frame anchoring.

The toed in or out condition is worse near the floor. The header prevents the upper portion of the jambs from toeing in or out but as you move away from the header towards the floor, the jambs have greater potential to be toed in or out during installation.

The only frame with this condition which can be corrected with little difficulty is the “slip-on” drywall frame. Since the “slip-on” drywall frame uses only compression anchors near the head and sill anchors near the bottom of the jambs, this condition is easily corrected.



Note that the rabbet surfaces are not 90° to the plane of the wall. This can be checked by placing a straight edge across the face of both jambs.

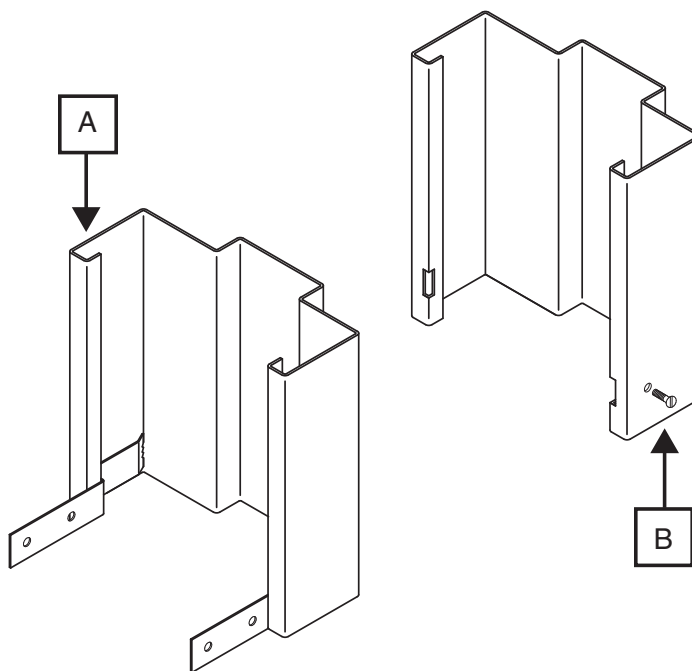
The straight edge should set flat across both faces at the same time. If the jambs are twisted, a “gap” will be created as shown in the figure to the left.

Typical toed-in or out condition

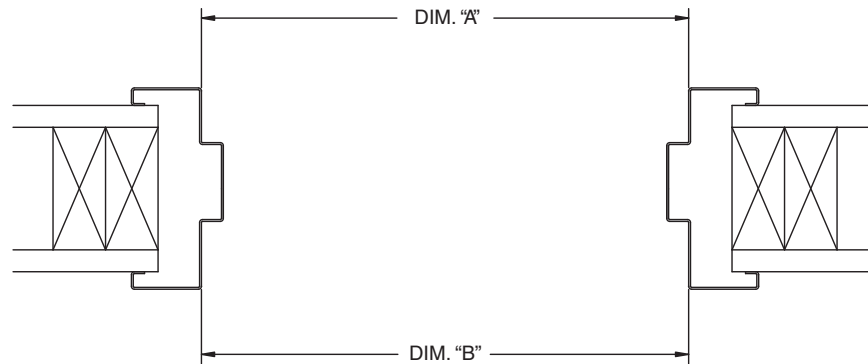
The figures to the right show the two typical types of sill anchors used on the “slip-on” drywall frame.

“A” shows an anchor which will be covered up by the baseboard trim. To gain access to this anchor the baseboard trim must first be removed. Next, the nails or screws holding the anchors to the wall must be removed. This will loosen the jamb and allow it to be squared in the opening.

“B” illustrates a screw going through the face of the frame into the wall. This holds the sill of the frame in position. This screw is exposed and is readily accessible. Removing the screw will loosen the jamb and allow it to be “squared-up” in the opening. If both jambs are toed in/out, they both should be “squared-up.” This can be checked by taking measurements as shown below. Both the “A” and “B” dimensions will be the same when the frames are properly set. However, the frame jambs must be plumb, level, and square with the head of the frame.



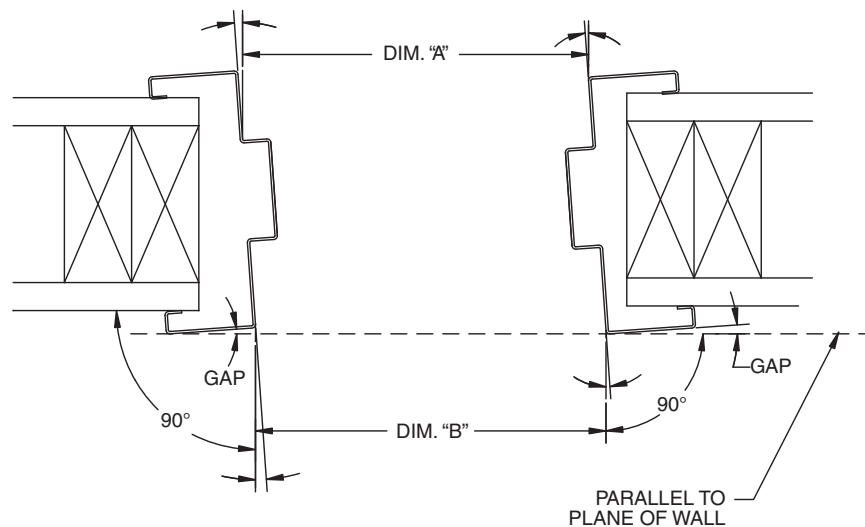
FRAME WITH RABBETS TOED IN OR OUT (continued)



If both jambs are toed in/out, they both should be “squared-up.” This can be checked by taking measurements as shown above. Both the “A” and “B” dimensions will be the same when the frames are properly set.

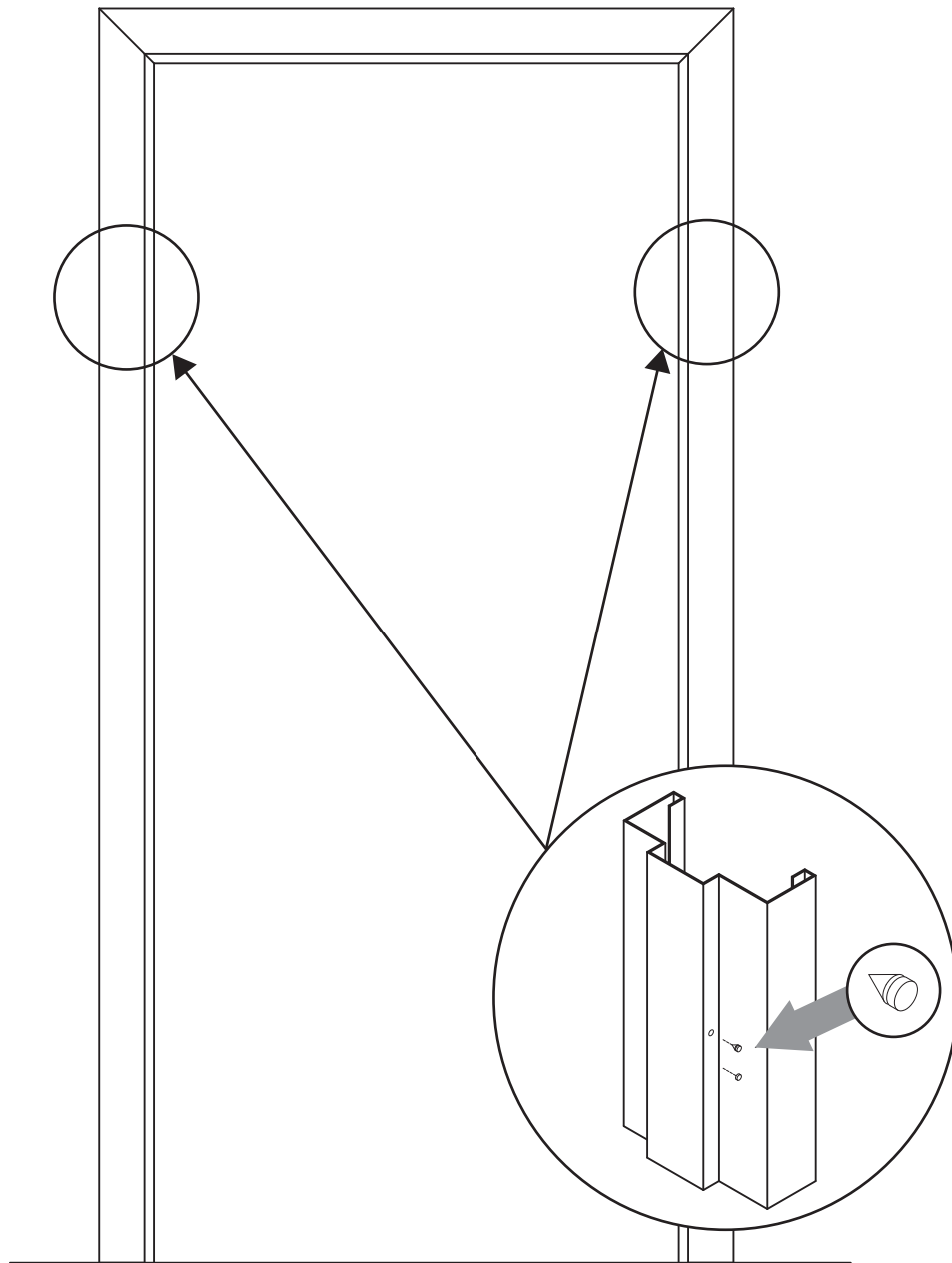
The below figure shows a condition which can give the impression that the frame is properly installed. Both dimensions “A” and “B” will be equal but the frame will not be square in the opening. Both jambs can be twisted to create a parallelogram. This can be checked by placing a straight edge across the face of both jambs (the straight edge is represented by the dotted line shown below). The straight edge should set flat across both faces at the same time. If the jambs are twisted, a “gap” will be created as shown in the illustration. The “gaps” will be to the same side if a parallelogram was created. The gaps could also be to opposite sides as shown in the Typical Toed In or Out Condition figure on the previous page.

This condition can be corrected as outlined for the other “toe out” or “toe in” examples.



SILENCERS

Some manufacturers provide a silencer hole in the frame stop to accept a push-in silencer. A “stick-on” silencer is also available for application to frames without the silencer hole. The silencer acts as a “rubber bumper” which evenly holds the door off the stop at a constant distance. If the door is held off of the stop rather than being allowed to move slightly towards the stop, the latch tube will fit tighter into the strike. Three silencers are placed on the strike jamb, one toward the top of the jamb (nearer the header) and one toward the bottom of the jamb (nearer the sill) and the third near the strike preparation.

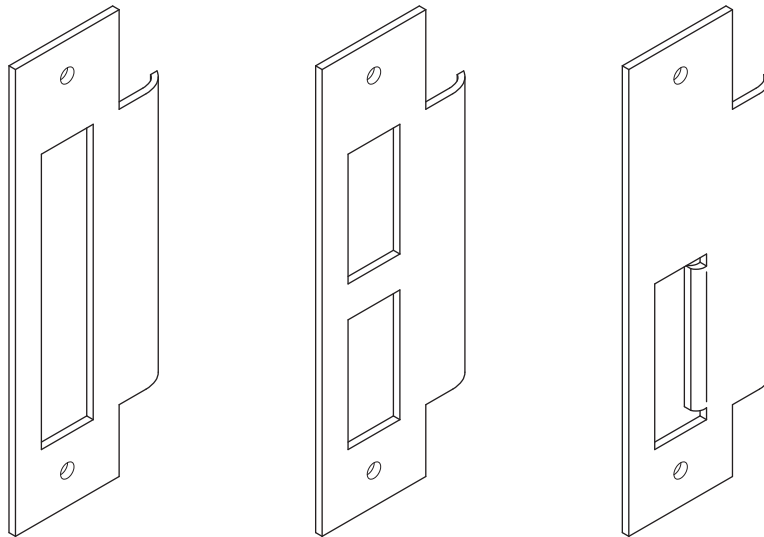


LOCK FITS LOOSE ON STRIKE

This condition cannot be caused by an improperly set door and/or frame. If the lockset on the door was not properly aligned with the strike on the frame, interference would occur or the latch bolt would bind in the strike. Since this is not the case; the cause is something other than improper installation. The following should be checked:

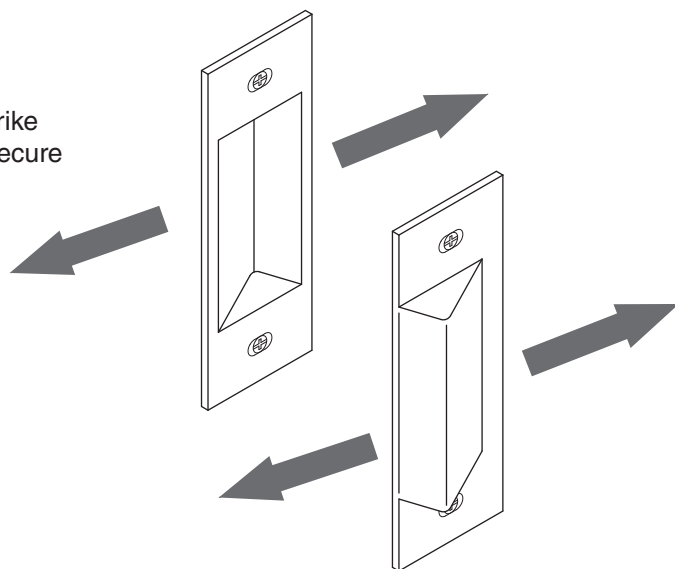
PROPER STRIKE

Is the proper strike plate attached to the frame?



RIM EXIT DEVICES

An adjustable strike plate is provided. Is the strike plate properly adjusted to provide positive and secure latching?



LOCK FITS TOO TIGHT IN STRIKE

Proper installation of the door and frame are essential. If the door and frame are not properly aligned, the latch bolt could bind in the strike. The following items should be checked:

IMPROPER CLEARANCE/DOOR SAG

For various conditions, refer to pages 1-2.

HINGE BIND

For various conditions of hinge bind, refer to page 7.

TWISTED DOOR

For various conditions of twisted doors, refer to page 9.

TWISTED FRAMES

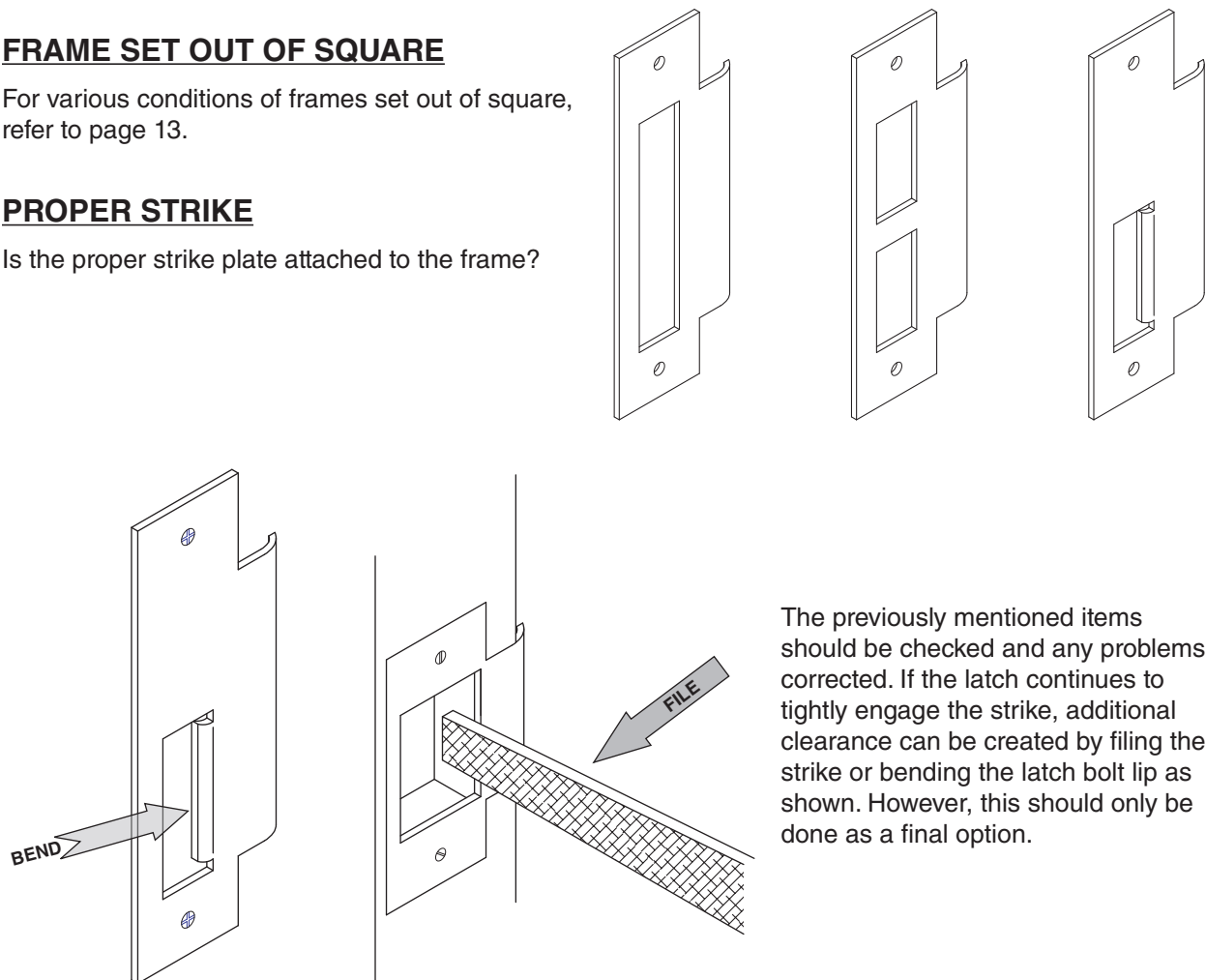
For various conditions of twisted frames, refer to page 12.

FRAME SET OUT OF SQUARE

For various conditions of frames set out of square, refer to page 13.

PROPER STRIKE

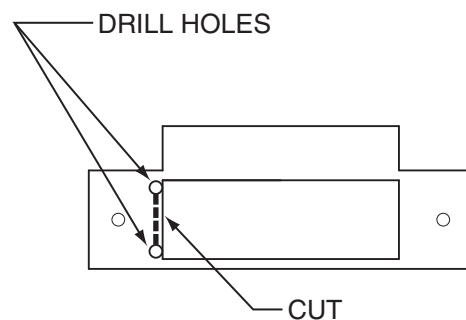
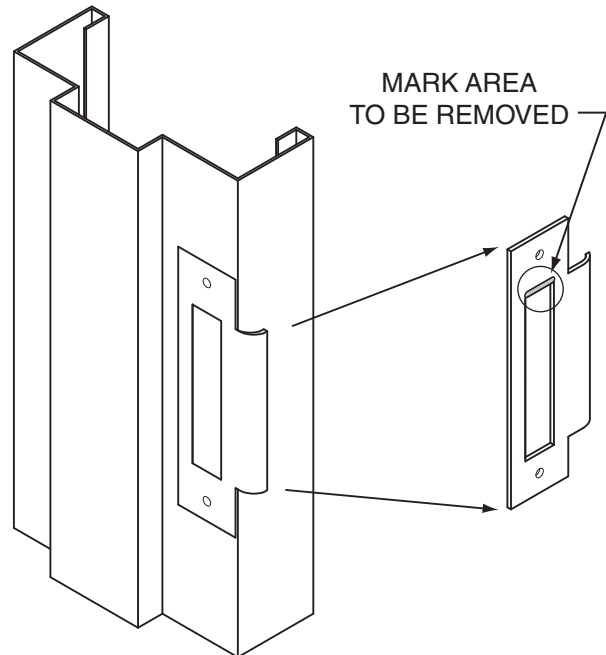
Is the proper strike plate attached to the frame?



LOCKSET OFF LOCATION ON DOOR OR STRIKE OFF LOCATION IN FRAME IN WALL

Minor location adjustments usually can be accomplished by altering the strike plate. The strike plate alteration would “extend” the opening to allow the latch tube to properly engage the strike. This is similar to the section “Lock Fits Too Tight in Strike” except that the amount of material removed from the strike is greater. This could compensate for minor mislocation of the strike plate on the frame and/or location of the lockset on the door. These general steps should be followed for modifying the strike plate.

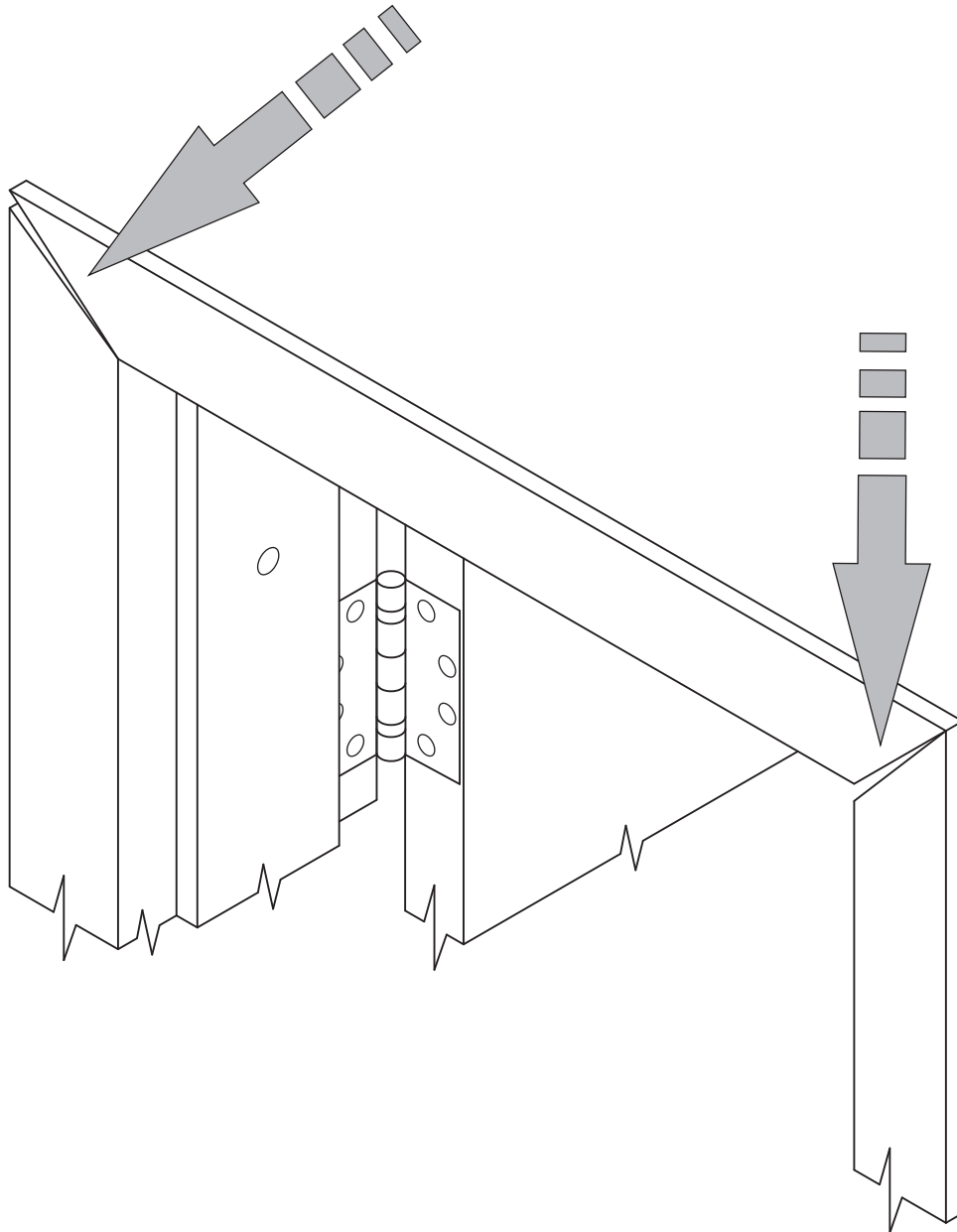
- Mark the area on the strike plate which must be removed.
- Remove strike plate from frame.
- Center punch corners of area to be removed and drill a $\frac{1}{8}$ " diameter hole in each corner.
- Use a saber saw and cut from the existing opening up to the $\frac{1}{8}$ " diameter holes. The remaining material between the two $\frac{1}{8}$ " holes should be filed away.
- Dress off all rough edges and corners with a file. This must be done since any irregular edges or edges with burrs could cause the latch tube to “drag” or “hang up” in the strike plate.
- Reassemble strike plate to frame and check alignment with lockset on door as well as latch tube function in and out of strike plate.



MITER NOT CLOSED ON FRAME

The major cause of miters not properly lining up and “closing” is incorrect installation. The effects of improper installation on miter seams are covered under the previous section of “Frame Set Out of Square.” This section should be referenced for detailed explanations of conditions which could occur, and solutions for correcting the problems.

This condition could also be caused by conditions as explained in the “Frame Loose on Drywall” section—figures on page 27.



CLEARANCE AT BOTTOM OF DOOR TOO SMALL

The available options for correcting this problem are limited. Frames which are permanently anchored to the wall construction (such as masonry or steel stud) leave few alternatives. The only available “fix” is to trim the bottom of the door, if the door’s construction will permit trimming. As shown in the figure below, the nominal “sill clearance” should be $\frac{3}{4}$ of an inch. The sill clearance is the dimension from the bottom of the door to the bottom of the frame jambs, and $\frac{3}{4}$ ” is a standard industry dimension.

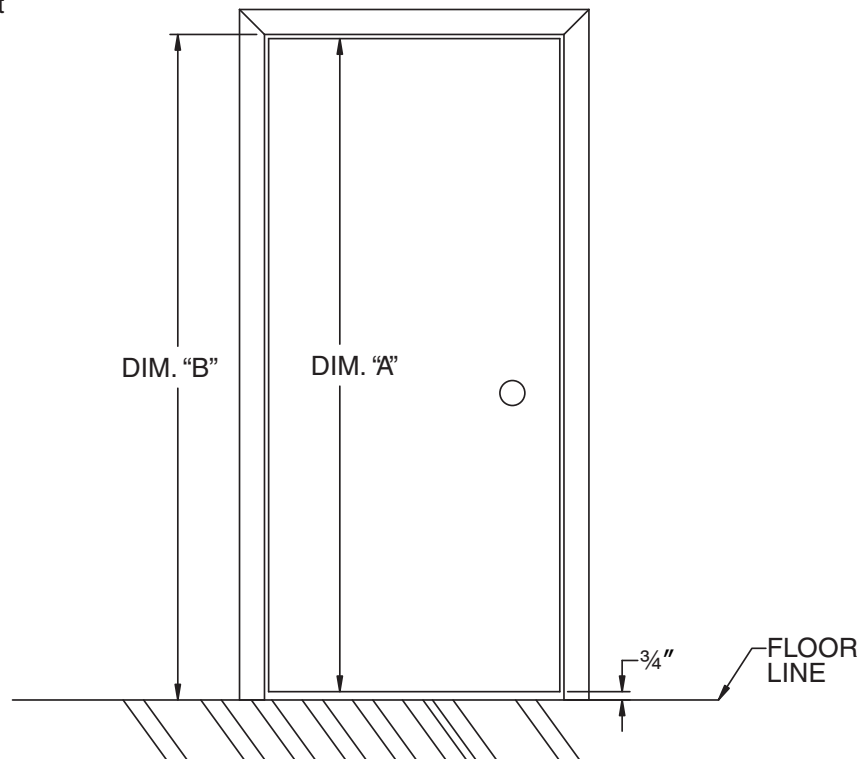
The frame jambs should be set at floor level which then results in $\frac{3}{4}$ of an inch between the bottom of the door and floor.

The $\frac{3}{4}$ ” also provides enough room to allow the installation of a “raised sill” which is a type of flat threshold. If a raised sill is used, the $\frac{3}{4}$ inch clearance is decreased.

Before any alterations are made the door height dimensions should be checked.

The door height is determined by taking a measurement represented by dimension “A” in the figure. If the door height is “oversize” the clearance at the bottom will be too small.

The frame jamb height (which is the same as the opening height) is determined by taking a measurement represented by dimension “B”. If the jamb height is “undersize,” the clearance at the bottom will be too small.



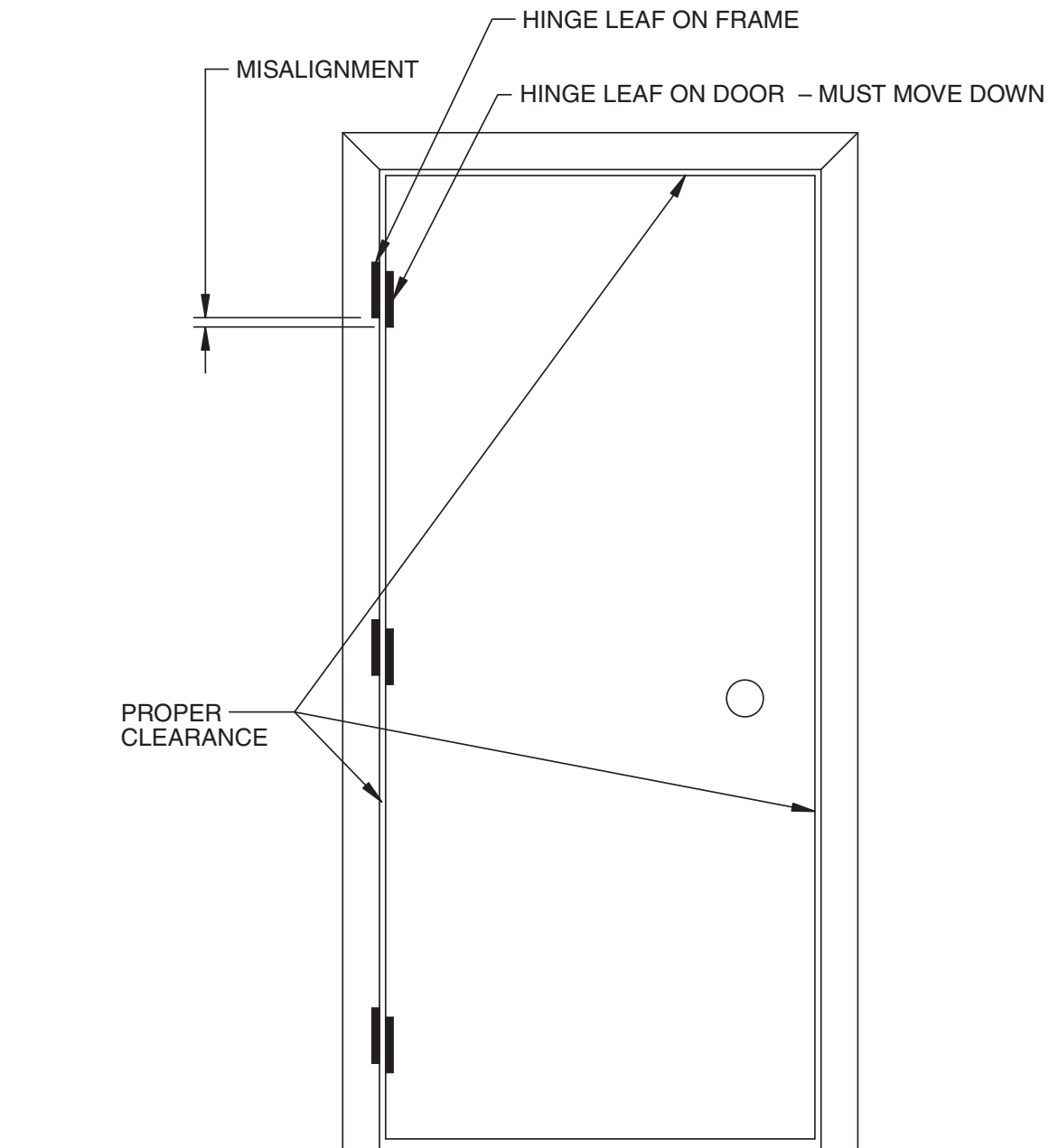
The bottom of the frame jambs should be set “on” the floor not “in” the floor. In some special cases the jambs can be set “in” the floor but this requires special design consideration and adjustments in door height and jamb heights. If this was not compensated for in the designed heights, the clearance at the bottom could be too small.

By taking these basic dimensions, you can confirm what element of the opening is in error. If trimming the bottom of the door is considered, you must make sure that the door’s design will allow such a modification to be made. It is best to contact the distributor or the manufacturer to determine if and how much the door can be trimmed. These types of trimming modifications can be relatively simple or more extensive depending on door design.

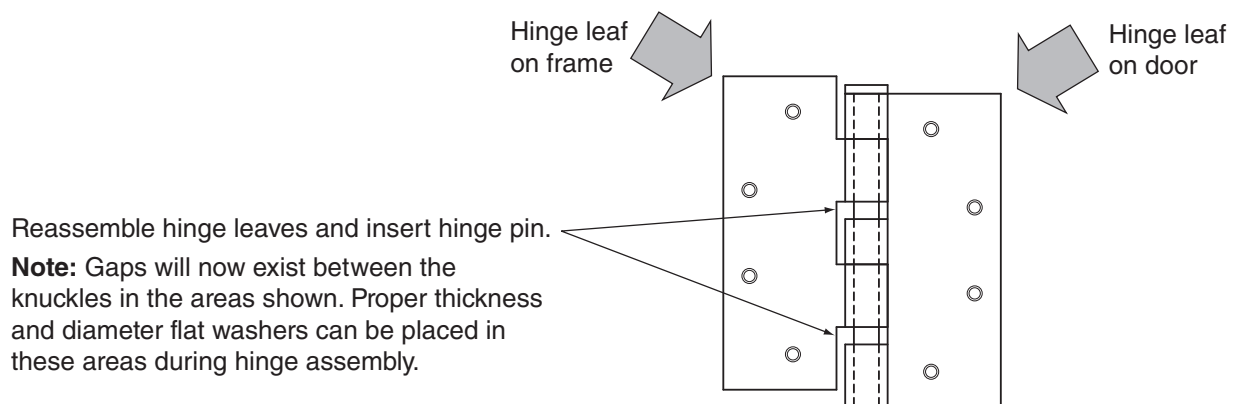
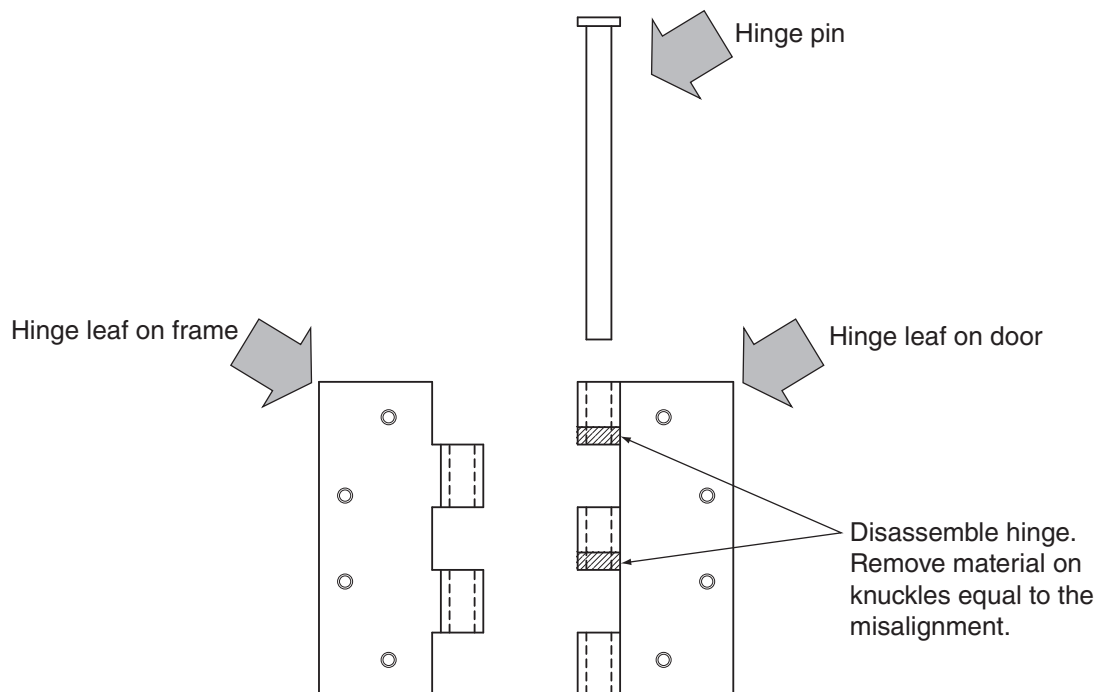
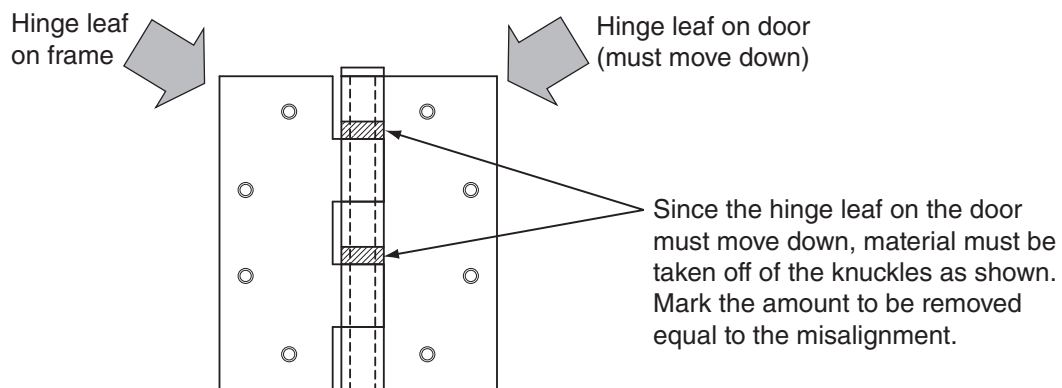
FRAME AND DOOR HINGE MISALIGNMENT

Frame and door hinge misalignment can cause a variety of problems. When misalignment occurs, either the door's hinge locations or frame's hinge locations are slightly off. This can apply to retrofitting existing openings where either the door or frame is being replaced, but not both. When new doors and frames are provided from a single manufacturer, this problem does not exist.

The example shown on the following page reflects the dropping of the hinge leaf which attaches to the door from its relationship to the hinge leaf on the frame. However, by removing material from the opposite end of the knuckles, the door hinge leaf could be raised as well. The frame hinge leaf can also be relocated in similar fashion.



FRAME AND DOOR HINGE MISALIGNMENT *(continued)*



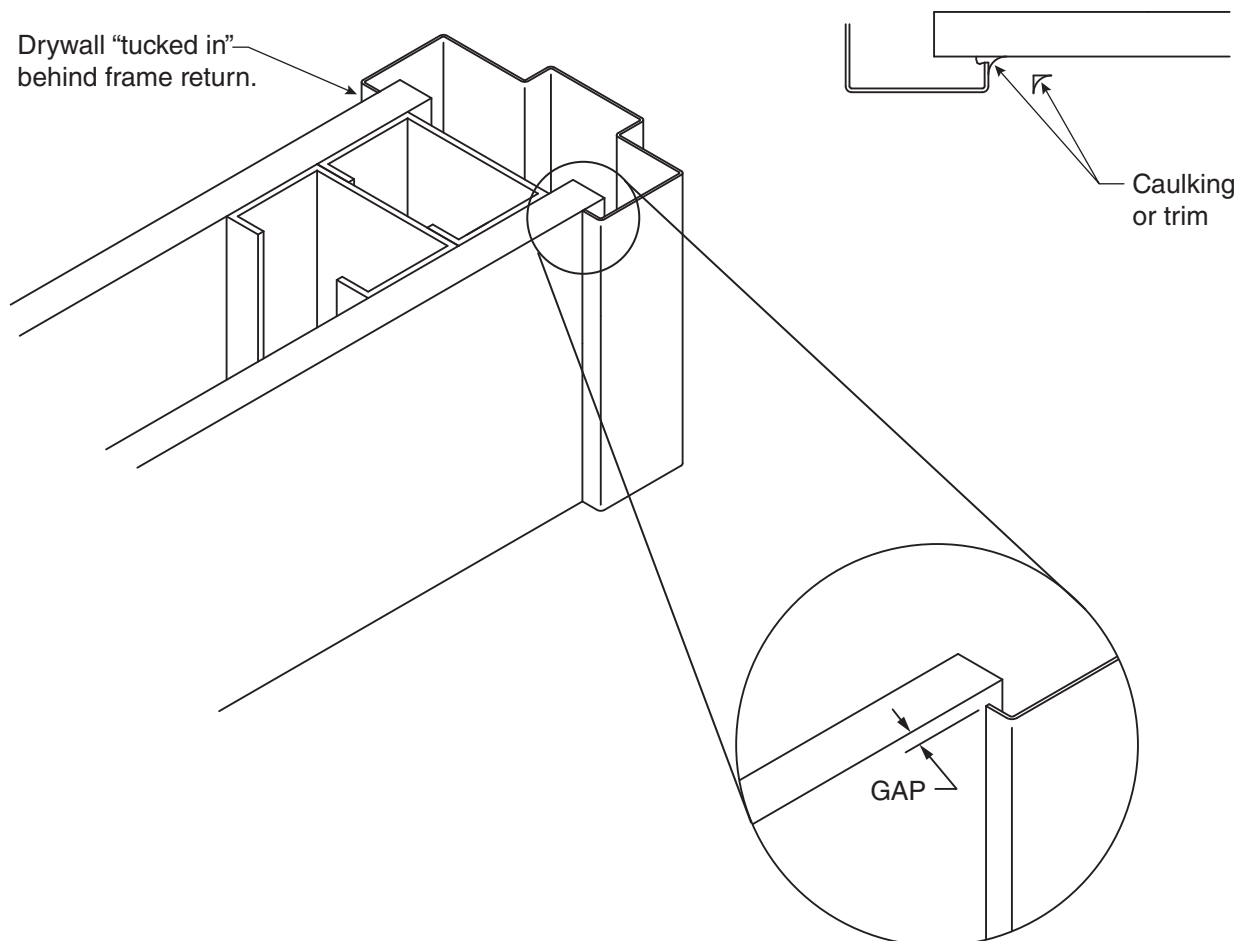
FRAME LOOSE ON DRYWALL

Frame manufacturers closely control the dimensions to which their frames are manufactured. Since automated equipment is used these dimensions are easily repeated from piece to piece. The majority of cases where frames are loose on drywall will reveal that the overall wall thickness has not been properly maintained. Wall thickness conditions can vary from undersize to oversize. The thickness should be checked, if possible, to verify the wall's compliance with the job specification.

Frames installed in drywall walls can use two different anchoring methods:

WELDED/SNAP-IN STEEL, OR WOOD STUD ANCHORS

Some frames use welded or snapped-in steel or wood stud anchors. These frames are installed prior to the drywall material being attached to the studs. The drywall can either be “butted-up” against the return of the frame or be “tucked in” behind the return of the frame. Only in the installation where the drywall is “tucked in” behind the return can there be a condition where the frame is loose on the drywall. This gap could be uniform along the entire length (height) of the jamb or could be only in certain areas. Since the frame cannot be removed, the only available options are to caulk the gap or cover it with trim.



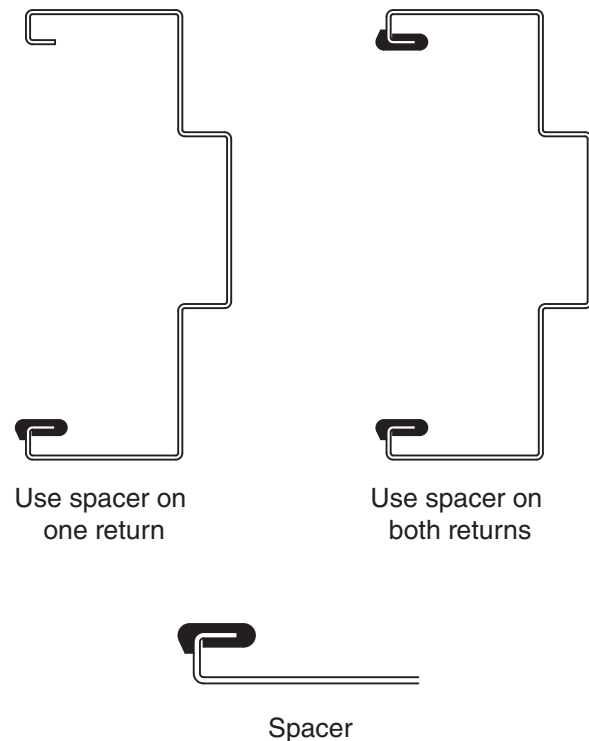
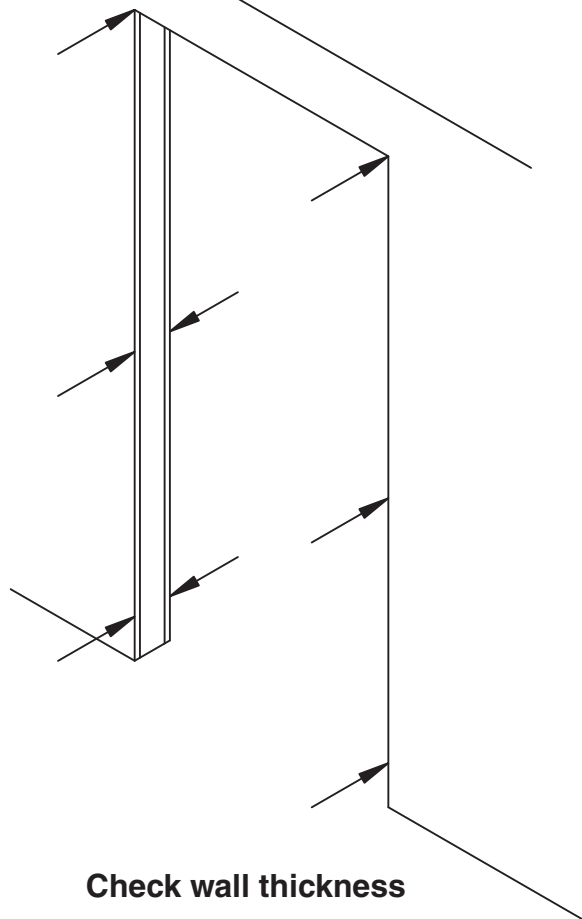
FRAME LOOSE ON DRYWALL *(continued)*

DRYWALL FRAMES WITH COMPRESSION ANCHORS

These frames are intended to be installed after the wall construction is complete. The anchoring methods that this type of frame uses allows the removal of the frame if so desired. The manufacturers' installation instructions should be followed anytime the frame is removed and reinstalled. Three conditions could exist for drywall slip-on frames which are "loose on the wall."

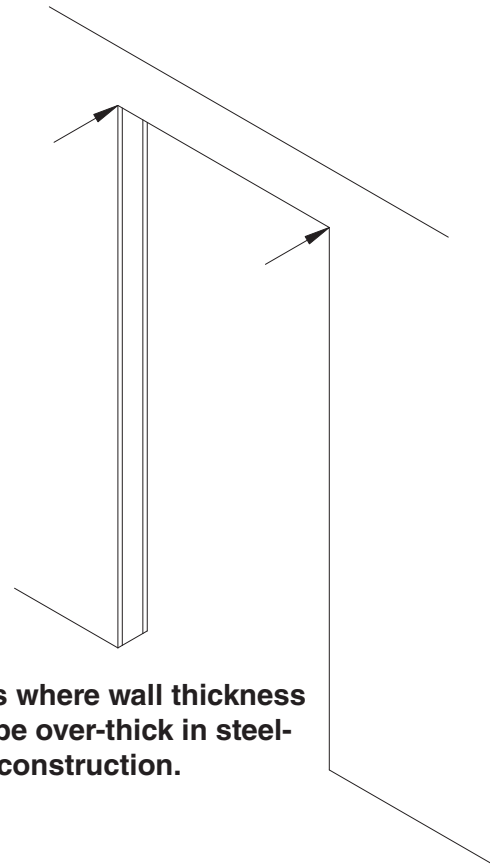
- 1) The first condition is a uniform gap along the entire length (height) of the frame jambs. The frame should be removed from the opening and the wall thickness checked at numerous places around the opening. These measurements should be compared to the job specifications. If the wall thickness is undersize, two options are available.
 - a) If the gap is relatively small, the use of caulking or trim can be considered to cover the gap when the frame is installed.
 - b) If the gap is larger, the use of "spacers" can offset the lacking wall thickness. The use of spacers requires that the frame be removed from the opening, the spacer(s) attached, and the frame reinstalled. The spacers can be used on either one or both of the frame returns which would result in spacers on either one side or both sides of the wall. Spacers are available from the frame manufacturer.

SPACERS ARE NOT TO BE USED ON LABELED FIRE DOOR FRAMES.

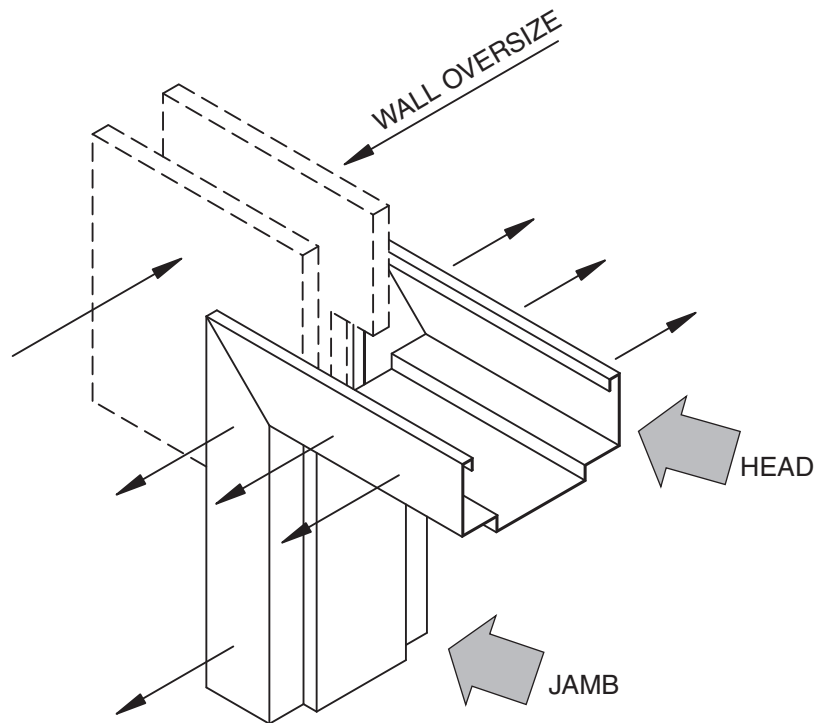


FRAME LOOSE ON DRYWALL *(continued)*

- 2) The second condition would be a gap that changes along the length (height) of the jamb. This condition is generally found in steel stud construction and results in “over thick” walls in the upper corners of the opening, refer to Figure 5. This is usually a direct result of how the steel stud headers were attached to the vertical steel studs to form the opening. When the wall is oversize (in this area) it will force the header and jamb miters to spread apart and actually open up the throat dimensions to accept the oversize wall. This will cause the corners of the frame to be extremely tight on the wall and as you progress down the jamb, a gap will begin to develop and then gradually disappear closer to the floor, see figure on following page.



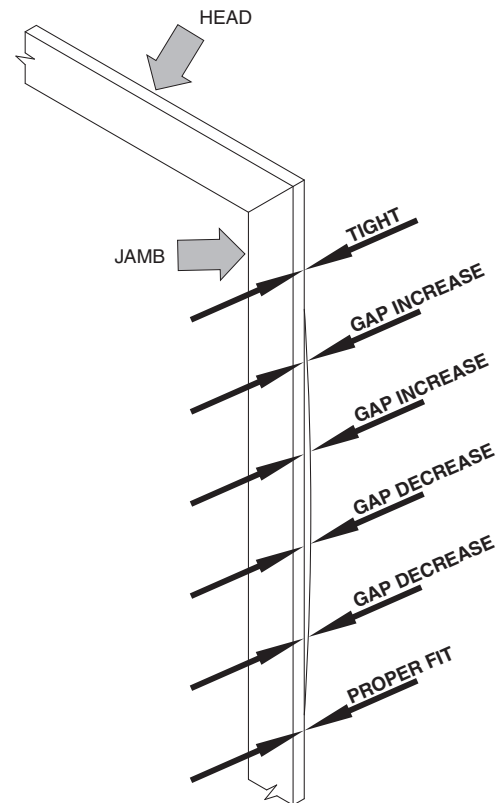
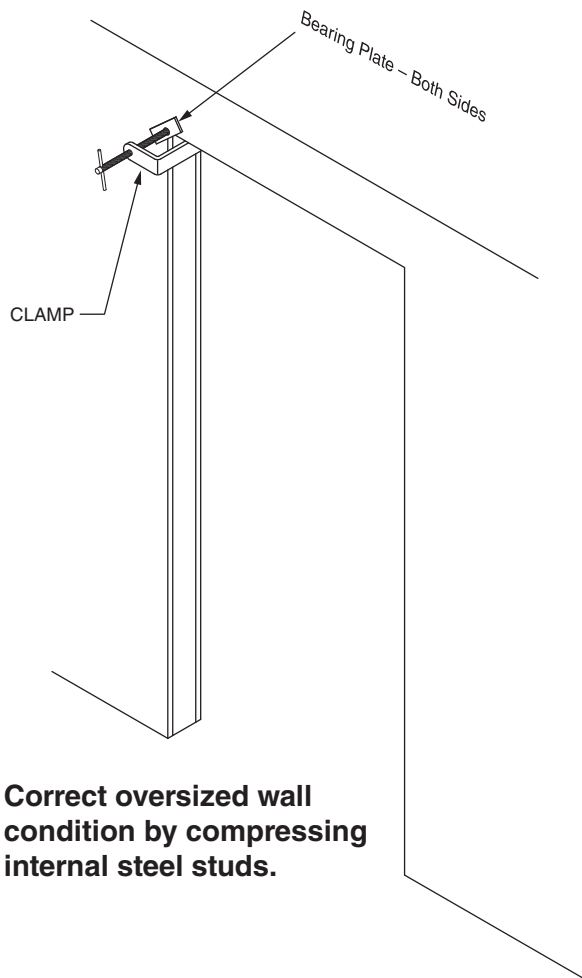
Areas where wall thickness may be over-thick in steel-stud construction.



Frame spreads apart with oversize wall.

FRAME LOOSE ON DRYWALL *(continued)*

This condition should be reported to the appropriate jobsite personnel. The condition can be corrected by putting a bearing plate on each side of the corner and compressing the internal steel studs with a clamp. However, the responsibility for correcting this condition belongs to the sub-contractor responsible for the actual wall construction.



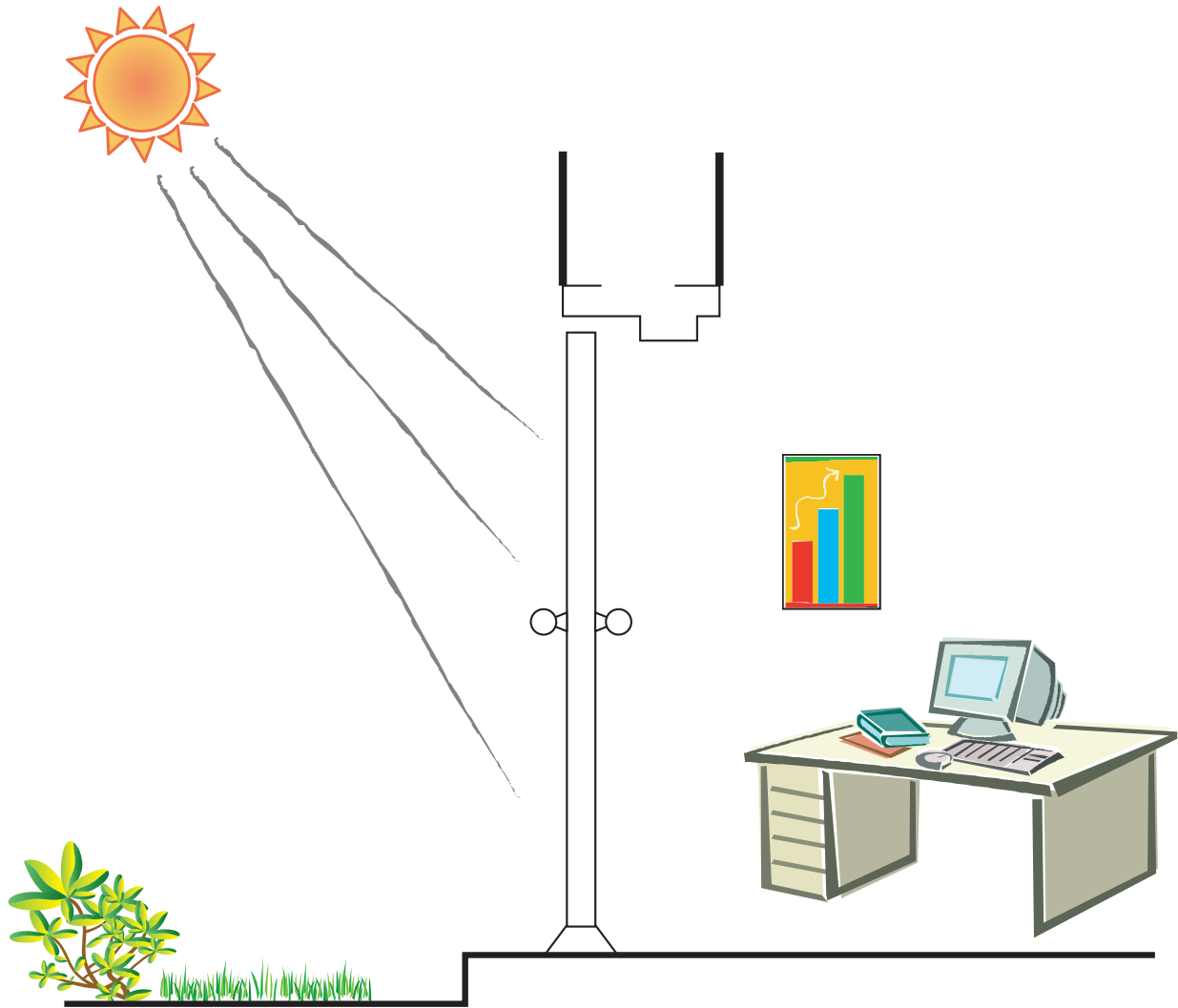
Gap produced by oversized wall.

- 3) The third condition is different from the first two which address the "fit" of the frame over the wall thickness. The cause of this condition is compression anchors which have not been tightened. The drywall frame would then be loose across the width of the opening and move from side to side against the rough opening.

The frame should be plumbed, squared and secured in the opening by properly adjusting the compression anchors following the manufacturers' instructions.

THERMAL BOW

Installers need to be aware of a condition known as Thermal Bow. Thermal Bow is a temporary condition which may occur in metal doors due to the inside-outside temperature differential. This is more common when the direct rays of the sun are on a door surface. This condition is temporary, and to a great extent depends on the door color, door construction, length of exposure, temperature, etc. This condition can often be alleviated by painting the exposed surface a light color. Thermal bow can occur in reverse under extremely cold conditions. Typical symptoms of thermal bow are hardware latching difficulty and door clearance issues.



METAL DOOR LTD